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Visualization and understanding of combustion processes using spatially and temporally resolved laser diagnostic techniques

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Abstract

Laser diagnostic techniques have for more than 30 years added very valuable input for a deepened understanding of combustion processes. The present paper will focus on techniques developed for visualization of important parameters with the ability to get detailed information in space and time. The paper is not meant to be a complete review of the entire research field but rather a survey with a majority of the contributions from the authors' laboratory. After a short introduction and background, essentially three techniques will be highlighted. Planar Laser-Induced Fluorescence, PLIF, will be described in terms of further development directed towards the use of a high-power Alexandrite laser, in some cases yielding a substantially increased sensitivity as compared to the more conventional Nd:YAG and Excimer lasers. There will also be a description of possibilities for high-speed visualization, 3D imaging and the potential to visualize species requiring a two-photon excitation process. Regarding thermometry, the paper will describe and exemplify the use of Thermographic Phosphors, mainly for surface measurements but also in droplets and gas flows. The third area to be described is the use of Polarization Spectroscopy, PS, for spatially and temporally resolved measurements in the IR spectral region, e.g. CO_2 and H_2O, various hydrocarbons, HCl and HCN. The last chapter focuses on present challenges and future applications. A major challenge is diagnostics in environments with limited optical access. Two methods that are applicable under such conditions are discussed, namely endoscopic approaches and picosecond LIDAR. Another challenging area addressed is application of diagnostics in optically dense environments, e.g. sprays, where a new technique, Structured Laser Illumination Planar Imaging, SLIPI, has been developed for suppressing the influence of multiple scattering. Finally, approaches for measurements of "new" species, challenges associated with LIF quantification, as well as measurements in environments possessing strong background radiation are discussed.
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1. Introduction and background

Despite its impressive age, combustion is still of crucial importance in many pivotal human activities, e.g. heating, transportation and waste handling, and it will remain so for years to come. The economies of many countries rely strongly on combustion for propulsion and power generation in industries producing cars, trucks and related products. Research, which will allow improvements in the efficiency and cleanliness of engines, gas turbines, furnaces,

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to optimize chemical processes, decrease the damage of fires, etc., will contribute significantly to competitiveness, sustainability and thus to the environment, economy and employment, and to the quality of life of people throughout the world.

In order to optimize and control combustion processes it is of crucial importance to have techniques which are able to measure critical parameters, such as temperature, species concentration, velocity, and particle characteristics. The challenges in this environment can be substantial; high temperature, high pressure, highly turbulent gas flows, and a variety of fuels, often involving phase transitions and thus multi-phase environments. Over the last few decades, perhaps the most important development for a deepened understanding of combustion processes, in terms of experimental achievements, has been the introduction of various laser diagnostic techniques.

The main advantages with these techniques are that they are non-intrusive, i.e. measurements can be made without introducing a physical probe that may influence measurements by altering the chemistry and/or fluid dynamics, and that measurements can be made with very high spatial ($\sim$10–50 μm) and temporal (<10 ns) resolution. Thanks to these features, laser techniques have been successfully applied in characterization of combustion/flow phenomena. Since the techniques covered in this review are based on high power lasers, which enable single shot two-dimensional visualization, the general drawbacks with these techniques are the need for rather sophisticated equipment and a relatively high cost. It should also be mentioned that the techniques described rely on short high peak power lasers where eye safety is an issue. Thus all measurements described require proper and adequate protection against any potential eye damage.

The main scientific directions in the area of development and application of laser diagnostic techniques for combustion studies may be divided into various categories according to:

1. Development of new diagnostic techniques as well as fundamental studies of both new and established techniques to increase their accuracy and precision. This category may involve rather fundamental aspects of atomic, molecular and laser physics, and nonlinear optics, and may not necessarily aim for a deepened understanding of combustion per se but rather the measurement techniques.
2. Applications of the more mature techniques for measurements of relevant parameters in laboratory flames for studies of various combustion phenomena, e.g. turbulent combustion, soot formation, and NO_x formation. These experiments often take place in close collaboration with experts from other areas, such as chemical kinetics and fluid mechanics.
3. Applications of mature techniques for characterization, optimization and control of industrial processes, i.e. in situ measurements in engines, gas turbines or furnaces.

The separations between these areas are not distinct and in reality large overlaps exist, relying on a close interaction of people and a flow of expertise and equipment between the different tasks.

The activities directed towards development and application of laser techniques for combustion studies has a long history and one of the first, if not the very first, conference in the field was arranged in Schenectady in 1974 [1]. This was the start of a rapid development, mainly in the first category above, involving the use of more powerful lasers and multidimensional detectors, new nonlinear optical techniques, e.g. [2–5] and multi-point imaging [6–8]. Already during the eighties there were thus invited talks on laser diagnostics at the Combustion Institute, e.g. [9–11]. During the following decades many applications for studies of combustion phenomena appeared, and now the focus has moved to category two and three as shown in invited talks at the Combustion Institute during the nineties, and the first decade of this century [12–14]. Today, the application of laser diagnostics in combustion processes is widespread. Aside from university activities, there are also governmental and industrial laboratories focusing on these areas all over the globe.

Various techniques have been developed and used for measurements of more or less all parameters of importance for characterization of combustion phenomena, i.e. species concentrations; molecules, radicals and atoms, temperatures; rotational, vibrational, translational and electron- temperatures, velocities, surface measurements and particle characterizations; sizes, number densities and volume fractions. The techniques which have been used can be split into spontaneous techniques based on linear optics, e.g. laser-induced fluorescence (LIF), Mie, Rayleigh and Raman scattering, where the signal is emitted in more or less all directions, i.e. 4π steradians solid angle, and coherent techniques based on nonlinear optics, e.g. Coherent anti-Stokes Raman Scattering, CARS, Polarization Spectroscopy, PS, and Degenerate Four-Wave Mixing, DFWM, where the signal is generated as a laser-like beam. For an overview of different techniques and their application in various combustion applications, in addition to the references above, we also refer to Refs. [15,16].

In the present paper we will present some activities representing categories 1–3 above. Due to space limitations, the main contributions will come from the authors' laboratory, but key references and contributions from other similar activities worldwide will also be given. The focus will be on applications of laser-induced fluorescence, including the use of a high-power, single-mode tunable

Alexandrite laser, two-photon LIF and high speed visualization, the development of a technique for spatially resolved absorption-based measurements in the IR spectral region, Polarization Spectroscopy, and the use of thermographic phosphors for temperature measurements, mainly on surfaces but also in droplets and gas flows. Special emphasis will also be put on special challenges requiring new techniques, e.g. measurements in combustion environments with very limited optical access, requiring single-ended experiments, and measurements in an environments where multiple scattering may limit the use of conventional techniques, e.g. in sprays. In the latter areas mainly two techniques will be emphasized, namely picosecond LIDAR and Structured Laser Illumination Planar Imaging, SLIPI, respectively.

2. Further developments and applications of Planar Laser-Induced Fluorescence, PLIF

Laser-induced fluorescence, LIF, is a widely used optical measurement technique in combustion diagnostics with the strength of imaging capacity, high sensitivity, high spatial resolution, species specificity and applicability in harsh environments.

LIF is based on laser excitation of atoms or molecules from a lower energy state, most often the ground state, to a higher excited state. Upon relaxation back to the lower state the atom/molecule re-emits radiation. This radiation can be analyzed and information on both temperature and concentration can be deduced. Since the excitation process is species dependent and the fluorescence from the upper state also occurs from, in the case of molecules, definite/discrete vibrational and rotational levels, the selectivity of the LIF technique is considerable. This selectivity combined with high sensitivity makes LIF a very powerful diagnostic tool. However, a major challenge with LIF is to convert the number of photons emitted to the number of molecules of the species of interest. The main problem is to account for de-excitation of the excited species through collisions of the excited species with other molecules/atoms, i.e. quenching. Since the number of species de-excited due to quenching is typically about three orders of magnitude larger than those giving rise to fluorescence, this loss mechanism is severe. Furthermore, since the quenching is dependent on colliding species, temperature and pressure, it is a delicate task to convert the detected LIF signal into number densities, especially for data taken in turbulent environment. Besides quenching there are a number of other barriers to quantitative measurements, such as linearity, signal attenuation and laser extinction. These issues are discussed in detail in [15,16] and references therein.

Planar Laser-Induced Fluorescence (PLIF) has since the early 1980s become one of the most powerful tools for studies of turbulent combustion phenomena, both for fundamental and applied research. Enormous research efforts have been made in developing different PLIF techniques for quantitative and precise measurements in various practical combustion processes. For overviews of PLIF applications in combustion research, see e.g. [15–18].

In the current presentation, rather than aiming at reviewing various applications, some further developments of the PLIF technique, involving their applications, are given below.

2.1. Use of an Alexandrite laser

Clearly, single-shot PLIF is one of the most important tools for the investigation of turbulent combustion, revealing the instantaneous flame structure to increase the detailed understanding of turbulent combustion and to visualize the combustion/flow fine structure in applied combustion apparatus, e.g. engines and gas turbines. For this purpose sufficient laser energy distributed in a plane is a necessity. So far in the history of PLIF almost all measurements have been performed using either a Nd:YAG or an Excimer laser, often in combination with a dye laser. However, one drawback with these systems is that the pulse duration is on the order of $\lesssim 15$ ns, which may result in excessive laser intensity, driving the molecular/atomic transition to saturation. In this case the fluorescence emission is limited and can not respond to an increase in laser power, which thus sets a limit on how strong the PLIF signal would be. To further increase the signal intensity, and to avoid saturation, it could be advantageous to increase the laser pulse length and/or the laser linewidth, in the latter case to excite more than one rotational line.

In order improve the signal-to-noise ratio in PLIF measurements the use of high power excitation laser pulses with longer duration and a broader linewidth to increase the PLIF signal intensity have been investigated and demonstrated. For this purpose an Alexandrite laser system [19] has been utilized which fulfilled all the properties mentioned above. The laser running at 10 Hz has a tunable wavelength covering 740–790 nm and a maximum pulse energy of $\sim$400 mJ. The laser can be operated either injection seeded for single-longitudinal mode or without seeding for multi-mode operation with a linewidth of about 8 cm^{-1} and a pulse duration of $\sim$100 ns. Detailed investigations on how this laser can be used for increased detectability of CH and HCO have been made. Both of these species are frequently used as flame front markers and CH is an intermediate in NO_x formation processes whereas HCO is used for spatial localization of maximum heat release, which makes them attractive for single shot two-dimensional imaging.

For CH PLIF detection, the transition X–B (0, 0) was adopted for excitation with the multi-mode laser frequency doubled to around 387 nm to cover several rotational lines in the molecular band head. Following excitation, the fluorescence from the B–X(0, 1) and A–X(1, 1) transitions around 430 nm were collected and detected with an ICCD camera. Around 70 mJ per pulse of the broadband 387 nm laser radiation was found to saturate the CH PLIF signal in an atmospheric pressure flame using a laser sheet with a height of 5 cm. Based on the laser specifications, a signal enhancement of about two orders of magnitude was estimated as compared to conventional Nd:YAG-pumped dye or OPO lasers, which enables the detection of CH down to sub ppm levels [20]. Shown in Fig. 1 is a single-shot PLIF image of CH in a highly turbulent jet flame when a laser sheet of 90 mm was used.

The high signal-to-noise ratio achievable with CH PLIF makes the technique a powerful tool for flame front structure investigations, especially for turbulent conditions where dramatic changes in local equivalence ratio occur, resulting in a high dynamic range of the CH concentration. This technique has also been applied together with a conventional OH LIF technique for simultaneous single-shot imaging of CH and OH radicals using PLIF to investigate partially premixed turbulent jet flames. Various flames have been stabilized on a coaxial jet flame burner consisting of an outer and an inner tube of diameter 22 and 2.2 mm, respectively [21]. From the outer tube a rich methane/air mixture was supplied at a relatively low flow velocity, while a jet of pure air was introduced from the inner one, resulting in a turbulent jet flame on top of a laminar pilot flame. The turbulence intensity was controlled by varying the inner jet flow speed from 0 up to 120 m/s, corresponding to a maximal Reynolds number of the inner jet airflow of 13,200. The CH/OH PLIF imaging clearly revealed the local structure of the studied flames. In the proximity of the burner, a two-layer reaction zone structure was identified where an inner zone characterized by strong CH signals has a typical structure of rich premixed flames. An outer reaction zone characterized by strong OH signals has a typical structure of a diffusion flame that oxidizes the intermediate fuels formed in the inner rich premixed flame. In the moderate-turbulence flow, the CH layers were thin closed surfaces extending over the entire flame, whereas the OH layers were much thicker. In the high intensity-turbulence flame, the CH layer remained thin until it vanished in the upper part of the flame, showing local extinction and reignition behavior. The single-shot PLIF images have been utilized to determine the flame surface density (FSD). In low and moderate-turbulence intensity cases, the FSDs determined from CH and OH agreed with each other, while in the highly turbulent case a locally broken CH layer was observed, leading to a significant difference in the FSD results determined via the OH and CH radicals. Furthermore, the mean values and the corresponding standard deviations of CH and OH radicals were extracted to provide statistical information about the flames, which can be used for validation of numerical calculations.

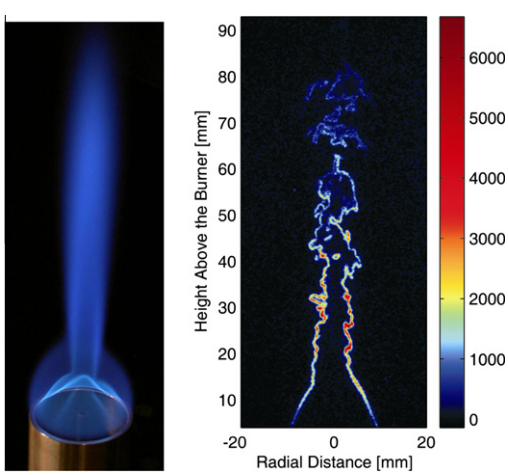

Fig. 1. Photograph of a piloted jet flame and a single-shot CH PLIF image taken from the cross-section through the symmetric axes.

HCO is another important radical present in most hydrocarbon flames. The same idea as described above for CH has also been used for improving HCO PLIF measurements, which potentially can be used as a generic flame front marker. Using the 3rd harmonic of the Alexandrite laser, a 10 mJ UV beam at ~258 nm was produced with 120 ns pulse duration and ~10 cm^{-1} laser linewidth. The application of such an excitation pulse has for the first time enabled single shot HCO PLIF measurements in turbulent jet flames [22]. Shown in Fig. 2 is a series of single shot HCO PLIF images taken in a premixed methane/oxygen flame burning on a 2 mm diameter welding torch burner.

2.2. Two-photon LIF

Other important species in combustion where PLIF measurements have been sparse, are for example light atoms; H, O, C, N and molecules like CO, H_2O, and NH_3, which have their absorption lines located in the VUV spectral region, i.e. <200 nm. Due to the strong absorption by air and the difficulties in generating tunable radiation with sufficient power in this wavelength regime, conventional one-photon LIF is not applicable. Instead, a two-photon excitation process can be used.

Two-photon LIF differs from single-photon LIF mainly in the following respects:

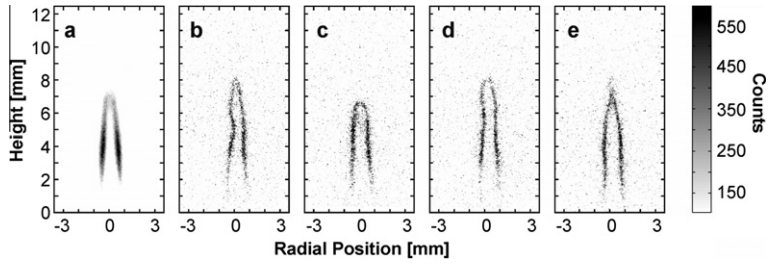

Fig. 2. HCO PLIF in a CH_4/O_2 welding torch flame: (a) 100 single-shot averaged image; (b–e) consecutively recorded single-shot images.

- different selection rules;
- Doppler-free measurements are possible;
- the temporal behavior of the excitation pulse is important;
- the LIF signal is proportional to the laser-intensity squared (ideally).

The last point is, from an experimental point of view, very important. This characteristic means that if a laser beam is focused in an environment with constant species concentration, the two-photon LIF signal will be highest in the focal region of the laser beam, in contrast to single-photon LIF where the signal intensity is independent of focusing since the increased laser intensity, I, is exactly balanced by the decreased probe area, A, i.e.

$$I_{\text{one-photon}} \sim I/A \times A = I,$$
$$I_{\text{two-photon}} \sim (I/A)^2 \times A = I^2/A \quad (1)$$

Despite its low transition probability, two-photon LIF was first demonstrated in the early eighties for pointwise detection of some species, e.g. O [3] and H [23]. Furthermore, PLIF measurements of these species have been demonstrated [24], even for engine conditions, e.g. H_2O [25] and CO [26]. Carbon monoxide, CO, is an important species, not only because it is a major pollutant, but it can also be used as an indicator of combustion efficiency, since incomplete oxidation of hydrocarbons results in increased concentration of CO in the burnt gas. Figure 3 shows single shot CO PLIF images recorded in a single-cylinder four-stroke side-valve SI engine at different crank-angle degrees [26]. The laser sheet was sent through the combustion chamber right above the valves where the spark plug was located at the lower right corner. Compared with flame chemiluminescence images, which were also recorded, it was found that although a strong flame chemiluminescence was observed at 30 CAD after TDC, the CO LIF had already decreased.

Clearly, quantification is an even more challenging issue for two-photon LIF, being a nonlinear technique, than for single-photon LIF. To achieve a quantitative interpretation of the two-photon LIF signal, substantial efforts must be put into measuring important parameters such as the two-photon absorption cross-sections, photo-ionization cross-sections, ac Stark shifts, collisional broadening and shift, and species-specific and temperature-dependent quenching cross-sections. Furthermore, many of the species requiring a two-photon excitation step are excited in the UV spectral region, e.g. H – 205 nm, N – 211 nm, O – 226 nm, and CO – 230 nm. To compensate for the weak cross-section of the two-photon excitation, high peak power lasers are commonly applied, which consequently may introduce strong interferences from fragments generated by photolysis, especially when species are to be probed in a combustion environment. For example photochemical effects from vibrationally hot CO_2 [27] has been investigated as a source of interference for two-photon LIF CO detection in atmospheric pressure flames. Also C_2 has been found to be a potential source of spectral interferences in detection of CO [28]. For the engine results displayed in Fig. 3, both the interference from C_2 and photolysis of hot CO_2 was found to be negligible based on spectrally and spatially resolved measurements.

In order to address the problem with photoproduction of species, and especially when detecting atomic species, it has been proposed to use a picosecond laser for excitation [29], considering the linear response of the photolysis production and nonlinear response of two-photon excitation. The application of picosecond laser pulses thus dramatically increases the peak power to keep the two-photon excitation efficiency high but decreases the laser pulse energy to reduce the photolysis interference. This technique has been applied for oxygen atom detection in flames providing interference free O concentrations [30,31]. Two-photon excitation of atomic hydrogen with a 205 nm laser beam was also investigated with combined ns and picosecond laser pulses [32,33] and the photolysis interference precursors have been investigated using the spatial distribution and concentration profiles [33].

2.3. High speed visualization

Until the late nineties many laser diagnostic combustion applications were made using

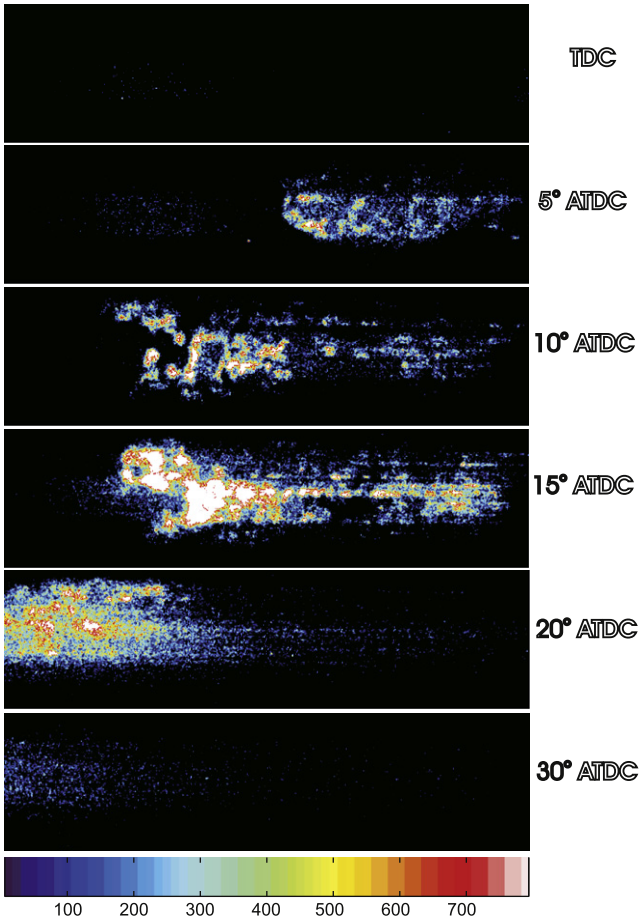

Fig. 3. Single shot two-photon LIF images of nascent CO from an SI engine recorded at different crank angles.

commercially available Nd:YAG lasers, which were limited to repetition rates in the range ~10–50 Hz. Such temporal resolution excludes the possibility of following relevant turbulent phenomena in real-time. For example in engine applications the limitation in repetition rates implies that only one image can be captured per cycle. Excimer lasers can be slightly faster but still not sufficient for temporally resolving turbulent time scales. Cu^{+} lasers can reach repetition rates of tens of kHz, but the low pulse energies make them suitable only for Mie scattering applications. The standard approach to circumvent the lack of sufficient temporal resolution, when following rapid events, e.g. in turbulent flames, is to measure probability density functions by capturing a sufficiently large amount of data to obtain adequate statistics. In the case of studying IC engines, e.g. spray development or flame propagation, the typical solution is to capture data from different cycles at subsequent crank angles. The recorded data are then put together in order to cover the entire event. If only average properties are of interest, the limitation of one image per cycle is acceptable. However, for processes with low cycle-to-cycle stability, the single image approach will suffer. One example of a process with low stability is the spatial distribution of combustion initiation and progression in Homogeneous Charge Compression Ignition (HCCI) engines. Other examples of unstable processes are ignition and early flame development in direct-injection spark-ignited engines (DISI) where it is essential that the stratified fuel charge is transported to the vicinity of the spark plug in the correct quantities and that it arrives at the time of ignition. This far from stable process shows a significantly increased number of misfires compared to conventional SI engines.

High-speed laser diagnostics may allow for resolving turbulent processes and for single-cycle-resolved engine measurements, in which single combustion events can be followed in time. Cycle-to-cycle variations of complex events can thus be studied. As individual structures can be

followed in time, the large-scale development of, e.g. fuel injection, ignition events, or flame propagation, can be visualized. This is crucial when trying to identify the origins of instabilities, e.g. in the mixture transport. For example in swirling flows and flames a speed on the order of a few hundred Hertz is sufficient whereas for many practical applications a visualization speed of 5–10 kHz is required. The latter would be the case to provide one image per crank-angle degree in IC engine diagnostics.

To realize the need for high speed visualization, essentially two approaches have been developed during the last decade, and these are briefly described below.

Nd: YAG laser cluster – high speed measurement: In order to perform high-speed diagnostics a sufficiently fast excitation source is needed. One approach to achieve this is to combine the output from several lasers. Such laser-clusters have been developed and are used extensively at Lund University. The laser cluster consists of four flash-lamp-pumped Nd:YAG lasers. Each laser consists of a Q-switched oscillator and a single amplifier. When the Q switch is switched twice during the flash-lamp discharge, each of the Nd:YAG lasers produces two pulses with a short time separation. The beams from the four lasers are spatially combined using a scheme based on frequency doubling and dichroic mirrors [34]. The laser beam from the first laser is converted from 1064 nm into 532 nm, with a second-harmonic-generation crystal. This green beam is then combined with the 1064-nm beam from the second laser using a dichroic mirror, reflecting 532 nm and transmitting 1064 nm. The spatially overlapped beams then pass through the second-harmonic-generation crystal of laser number two. The 532-nm beam passes the crystal unaffected while the 1064-nm beam is converted into 532 nm. The same combining procedure is then repeated until all beams are spatially overlapped. The state-of-the-art multi-YAG laser (THALES) thus produces eight pulses with a pulse energy of ~400 mJ/pulse (or four pulses with ~1000 mJ/pulse) at 532 nm. After pumping a dye laser/OPO and frequency doubling, up to ~7 mJ of tunable radiation around 280 nm could be achieved in each of the eight pulses. The minimum time separation between the eight pulses is ~6 μs (170 kHz). In order to achieve 2D LIF images, a framing camera consisting of eight separate image intensified CCD detectors have been used (Hadland, or Specialised Imaging). The equipment above has been used in various applications ranging from more basic understanding of turbulent combustion [35] to IC engine applications [36].

One example where the multi-YAG cluster/Framing camera was used to perform high-speed visualization was the first published single-cycle-resolved fuel tracer PLIF imaging performed in an HCCI engine [37]. When running a Homogeneous Charge Compression Ignition (HCCI) engine the fuel and air are allowed to mix prior to combustion in order to create a homogeneous, or close to homogeneous, charge. The charge is then compressed until it auto-ignites. The following combustion occurs through distributed reactions throughout the entire bulk volume, resulting in a very high overall combustion rate. In Fig. 4, a series of PLIF images is shown. It can be clearly seen how ignition is initiated at multiple locations simultaneously. The fuel used in these measurements was ethanol. Due to the absence of fluorescence from the fuel itself, acetone, at a concentration of 10%, was added as tracer and served as a marker of unburned regions. The excitation wavelength from the YAG-cluster was 266 nm, and fluorescence was emitted in the spectral region between 350 and 550 nm with a peak around 430 nm. The laser energy was approximately 25–30 mJ in each of the eight pulses. In this HCCI engine, the charge is completely evaporated and well mixed before ignition occurs. This feature makes the matching of the vaporization characteristics for the fuel and the tracer less critical. Additional important characteristics are that the tracer has suitable fluorescent properties and that it has minimum influence to the combustion process itself [38].

Nd: YAG laser cluster – 3D visualization: Since essentially all practical combustion takes place under various degrees of turbulence, which is truly three dimensional, there is a need for 3D visualization. The equipment above has also been demonstrated and applied for 3D visualization of species distributions by rapidly sweeping the laser beams through the volume of interest. Fast 3-D measurements based on this principle have also been demonstrated using a flash-lamp-pumped dye laser [39]. A drawback with this approach is that the long laser pulse duration makes frequency doubling difficult to achieve which limits the energy available per exposure. The first demonstration of 3-D measurements with multi-YAG cluster/Framing camera was measurements of soot volume fractions in a turbulent flame [40]. To illustrate the technique an example of 3D fuel tracer visualization in an HCCI engine is shown in Fig. 5. The eight laser beams from the Nd:YAG laser cluster, fired with 10 μs temporal separation, are spatially separated using an oscillating mirror. This results in a spatial separation of 0.5 mm between the individual laser sheets within the combustion chamber. All data was collected in 70 μs, corresponding to 0.5 crank-angle degrees (CAD) at 1200 rpm. During this short acquisition time, the fuel conversion appears nearly frozen. To verify this assumption, images were recorded with the same time separation but with all eight sheets overlapped at the same spatial position in the engine. The cross-section of the laser sheet was $50 \times 0.25\ \mathrm{mm}^2$. The sheet was formed by a cylindrical lens ($f = -100$ mm) and a

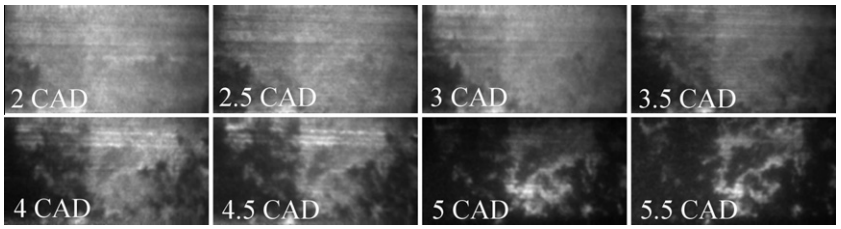

Fig. 4. A sequence of eight single-shot PLIF images showing the distributed reactions in an HCCI engine running on ethanol and acetone as fluorescent tracer [37].

spherical lens ($f = +1000$ mm). To achieve parallel laser sheets in the engine, the oscillating mirror was placed at the focal point of the spherical lens. The fluorescence was detected from beneath, through the piston extension. The camera was equipped with proper filters to transmit the LIF signal with a maximum at 430 nm while discriminating against scattered laser light at 266 nm and rejecting fluorescence at longer wavelengths from oil residues on the cylinder surfaces. The depth of field of the imaging optics exceeded the dimensions of the scanned volume. Examples of the eight individual PLIF images are shown in Fig. 5.

A number of image processing steps have been applied to the acquired PLIF images in order to compensate for background, absorption along the beam path, heterogeneities in the laser-intensity profiles and pulse-to-pulse energy fluctuations. To visualize the sequences as three-dimensional surfaces a shape-based interpolation scheme was used. The initial step in this procedure is to produce iso-concentration curves from the eight two-dimensional images by choosing a suitable threshold. Shape-based interpolation is then used to create new iso-concentration curves, in this case seven, between the original (measured) image planes. From these iso-concentration curves, a three-dimensional surface is constructed and examples are shown in Fig. 5d and e. For a more detailed description see Ref. [41].

Laser visualization using kHz lasers: During the last few years a complement to the Nd:YAG laser cluster approach for high speed visualization has been developed. In this case a diode-pumped Nd:YAG (or YLF or YVO_4) laser allows repetition rates up to tens of kHz. The specifications of a state-of-the-art laser of this kind (EdgeWave HD 40IV-E) show maximum pulse energies at 532, 355, and 266 nm of, 12, 5 and 3 mJ, respectively, at a repetition rate of 10 kHz. There is also a possibility to pump a specially designed dye laser providing, at the frequency doubled wavelength required to excite the OH molecules at around 283 nm, around 200 μJ at 1.5 kHz [42]. This approach has been

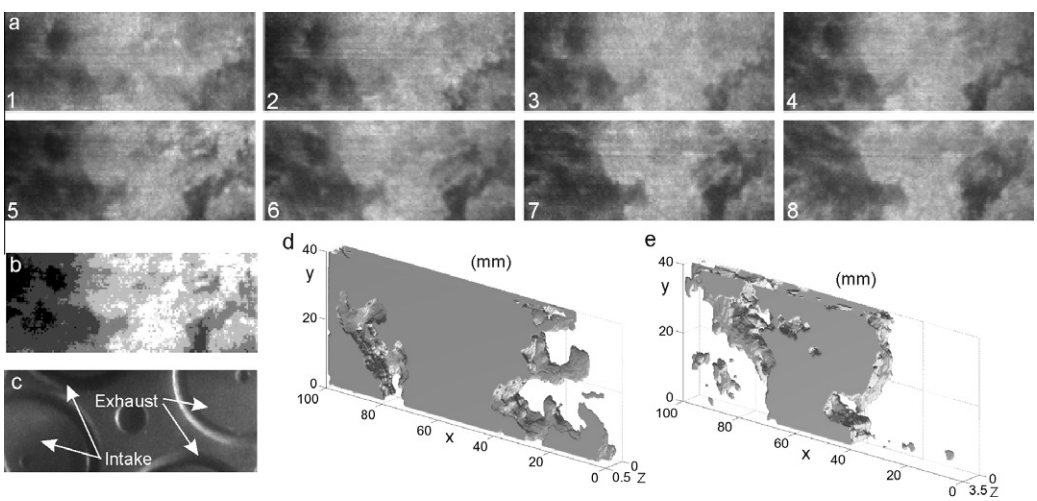

Fig. 5. (a) A three-dimensional fuel tracer LIF sequence. Eight parallel and equidistant 2D cuts of the fuel distribution in the HCCI engine are shown. The data were recorded at 2CAD after TDC. (b) This image shows the five different grey scales corresponding to LIF intensities of <20%, 20–40%, 40–60%, 60–80% and > 80% of the maximum intensities in frame 5. (c) Field of view for detection through the piston. (d) Three-dimensional fuel iso-concentration surface, corresponding to 50% of the maximum LIF intensity, calculated from the data presented here, $z = 0$ at the first frame. (e) Three-dimensional fuel iso-concentration surface corresponding to 70% of the maximum LIF intensity [41].

successfully applied for high speed visualization of combustion phenomena by several groups (e.g. [43,44]). In these cases high speed CMOS detectors, capable of capturing thousands of images at corresponding frame rates, have been used for recording of the LIF images.

As an example, Peterson and Sick performed simultaneous application of particle image velocimetry (PIV) and Planar Laser-Induced Fluorescence (PLIF) of biacetyl, used as a fuel tracer, at a rate of 4.8 kHz in a spray-guided spark-ignited direct-injected (SG-SIDI) engine [43]. The aim of their study was to understand the in-cylinder conditions that exist in the vicinity of the spark plug during ignition. An Nd:YLF (Quantronix Darwin Duo) dual cavity laser producing 5.9 mJ/pulse at 527 nm was used for the PIV measurements. An Nd:YAG (Quantronix Hawk I) laser delivering 1.4 mJ/pulse at 355 nm was used as excitation source for the fuel visualization. A 12-bit CMOS camera (Phantom V7.1) detected the PIV/Mie scattering and a 14-bit CMOS camera (Phantom V7.3) equipped with a lens-coupled image intensifier (LaVision HS-IRO) captured the weaker biacetyl fluorescence. Simultaneous flow field and fuel concentration measurements proved to be feasible with single crank-angle degree resolution in the optical SG-SIDI (Spray Guided Spark Ignited Direct Injected) engine, operating at 800 rpm. A short series of four consecutive images is shown in Fig. 6 [43]. In this sequence it can be seen how the fuel spray plume impacts the spark plug. Part of the fuel passes through the spark gap, while another portion of the fuel impacts the ground strap and is reflected downward. Velocity gradient information was also extracted from the flow field measurements to gain knowledge about shear strain rate and vorticity in this critical area. In short, the measurements visualize instantaneously the distribution of equivalence ratio, velocity magnitude, shear strain rate, and vorticity values which varies substantially from cycle-to-cycle during the spray event. Such results could not have been achieved without high-speed diagnostics.

Another example of combined high-speed measurements using kHz lasers is presented by Heeger et al. [44] who performed two-dimensional high-speed OH-PLIF and PIV measurements in a turbulent non-premixed edge flame in an opposed-jet burner. The aim of the study was to investigate spark ignition in a turbulent non-premixed environment. Such processes are relevant, e.g. for relighting of gas turbine combustors. Mie scattering from aerosol droplets, was used for the high-speed PIV measurements. Excitation was provided from two diode-pumped frequency-doubled Nd:YVO4 lasers (EdgeWave, IS4II-DE) that could generate double-pulses. The simultaneous PIV/OH-PLIF experiments were performed at a repetition rate of 2.5 kHz. For the OH-PLIF measurements, a frequency doubled (523 nm) diode-pumped Nd:YLF laser (EdgeWave, IS 8II-E) was operated at 2.5 kHz (3.4 mJ/pulse) to pump a tuneable dye laser. The dye laser was a 10 Hz system (Radiant Dyes, NarrowScan) modified for high-speed pumping. The maximum output pulse energy achieved at 2.5 kHz was 48 μJ around 283 nm.

Also in this study CMOS cameras (LaVision, HSS5) were used for signal detection. As for the biacetyl PLIF mentioned earlier, the PLIF camera had a lens-coupled two-stage image intensifier (LaVision HS-IRO). After the spark, a spherical flame kernel with a diameter of about 3 mm was created. The kernel then formed a flame edge that spread radially, probably along the stoichiometric mixture fraction iso-line, as a sheet about 1.5 mm thick until it filled the whole stagnation plane between the two nozzles of the counter flow burner. Figure 7 shows simultaneous OH PLIF/Mie images of a successful ignition event. These images show an apparent difference between the thickness indicated by the OH-PLIF images and the corresponding thickness of the evaporated droplet region in the Mie images. This difference is due to the droplet evaporation in the preheat zone.

More state-of-the-art work using the kHz-laser approach can be found in [45–48]. Clearly, the optimum approach for high speed visualization, either a few high power pulses as in the Multi-YAG cluster approach, or the kHz laser, depends on the measurement situation, the species to be measured, the size of the area to be visualized and the speed required. For fuel visualization, it seems that the available pulse energy of the kHz system may be sufficient; the same may be true for OH, probably with the exception for high pressure applications. For some other species, e.g. NO and CH, the Multi-YAG approach may be advantageous. Furthermore, if a large area and a very high speed situation is investigated, the Multi-YAG will have an advantage. On the other hand if phenomena like extinction or reignition in stationary flames are to be captured, the limited number of pulses and lack of proper triggering of the Multi-YAG will favor the kHz approach.

Interestingly, there are also high speed lasers which may combine the two concepts, providing both assets by forming a larger burst, up to hundreds of high power pulses [49,50], which may be very attractive, although so far these systems are not yet commercially available. In some cases it may even be possible to utilize continues-wave (cw) laser excitation, as demonstrated recently by Cheung and Hanson [51].

3. Thermographic phosphors

Temperature is one of the most frequently required parameters for measurements in a wide

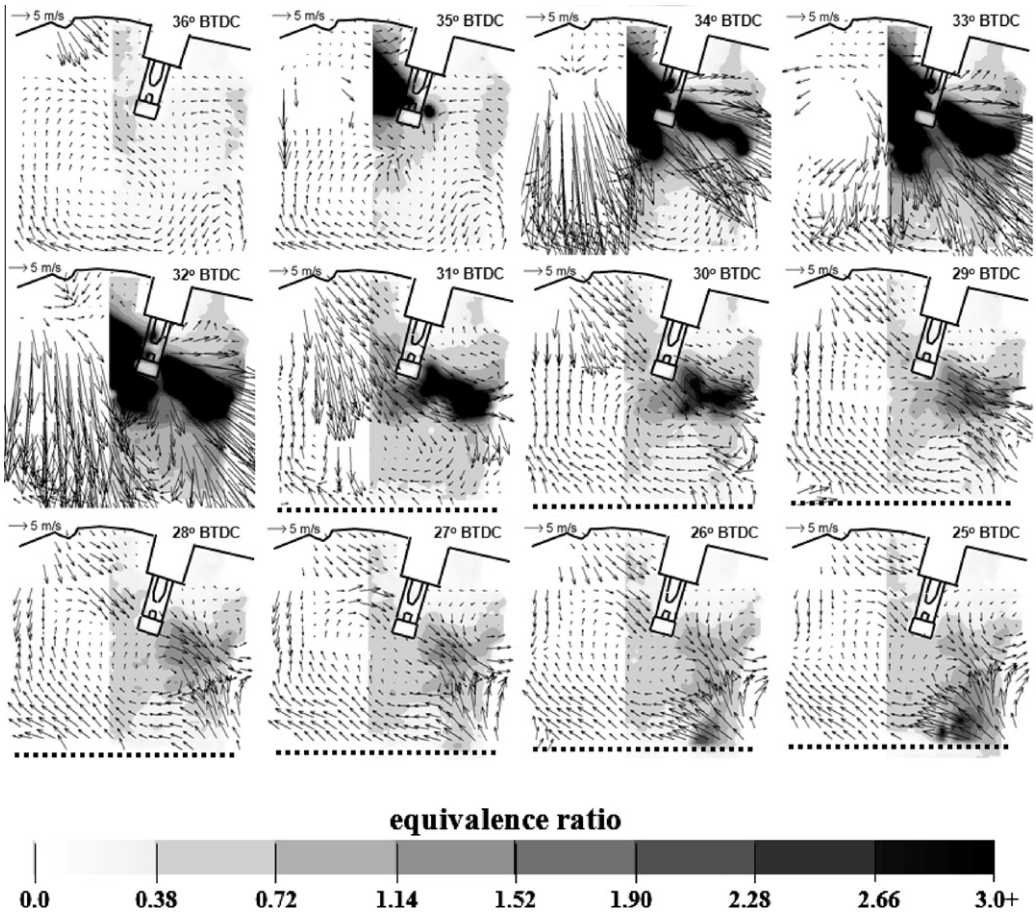

Fig. 6. Simultaneous fuel concentration and velocity measurements illustrate the strong coupling of spray momentum and air–fuel mixing inside a spray-guided DISI engine. (Courtesy of Peterson and Sick [43].)

variety of combustion applications. Besides measurements of gas temperatures, there is also a high demand for accurate measurements of surface temperatures since this information may lead to extended and optimized material durability and understanding of heat transfer phenomena which are of crucial importance for various combustion apparatus, e.g. IC engines, gas turbines and furnaces.

Traditional techniques for measuring surface temperatures include those relying on thermocouples or pyrometry. However, thermocouples in combustion applications are not only intrusive but also prone to a multitude of possible errors, including those of radiation, conduction along the wires, catalytic effects and the slow response. Pyrometry is indeed a non-intrusive technique based on measuring the naturally emitted electromagnetic radiation by all objects. However, in combustion applications emissivity variations, spectral interferences stemming from walls and flames and absorption by gases and soot are inevitable, making the results questionable. The present chapter describes research directed towards an alternative measurement technique based on the use of temperature-sensitive phosphors known as Thermographic Phosphors, TP, for temperature measurements in various combustion environments as well as in related areas, e.g. sprays.

Thermographic phosphors are composed of a carrier material doped with some activator material. The activator is often a rare-earth metal. The doping concentration is typically around one percent, which is low enough for the activator atoms to be isolated from one another by the host matrix. The host material is microcrystalline and mostly transparent to visible radiation, i.e. it is usually the activators that absorb and emit radiation. TPs have traditionally been used in everyday applications such as fluorescent-tube lamps, CRT–television sets and fluorescent paint. Although the temperature dependence of phosphors was discovered in the 1950s, it was not until the early 1980s when Allison and Gillies discovered the advantages

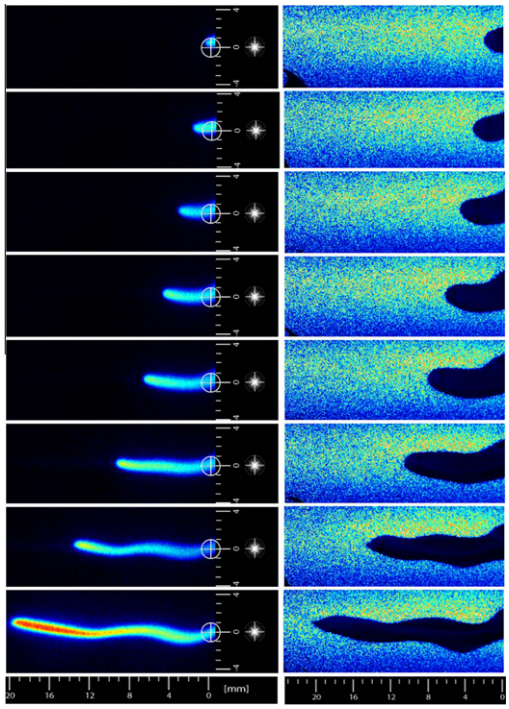

Fig. 7. Simultaneous OH-PLIF (left) and Mie (right) images captured in an opposed-jet burner. The images are recorded with a frame rate of 2.5 kHz and illustrate one single ignition event. The spark is positioned at $r = 4$, $z = 0$. Imaged area is 25×8 mm. (Courtesy of Heeger, Böhm, Ahmed, Gordon, Boxx, Meier, Dreizler and Mastorakos [44]).

of using thermographic phosphors for remote surface temperature measurements, see [52] for an overview of TP applications up to the mid-nineties.

The luminescence properties of thermographic phosphors are very sensitive to the variations in temperature. Most thermographic phosphors exhibit a broadband absorption in the UV region and few more narrow absorption lines in the visible region. Several methods can be used for excitation, such as laser radiation (pulsed or continuous), particle beams, chemical reactions or thermal excitation in flames. For combustion applications, the excitation is usually performed by an ultraviolet or a visible laser beam. Following excitation, the subsequent emission due to the relaxation of electrons from the excited state to the ground state can be used for thermometry. Two methods for inferring the temperature can be employed, the temporal and the spectral method. Both methods allow single-point or two-dimensional measurements to be performed. In the first case, illustrated in Fig 8a, the intensity decays exponentially and is in the simplest case single exponential in accordance with

$$I = I_0 e^{-\frac{t}{\tau}}, \tag{2}$$

where I_0 is the initial emission intensity, t is time and T is the lifetime time constant of the phosphorescence involved, i.e. the time taken for the intensity to decrease to $1/e$ of the initial emission I_0

For some thermographic phosphors the intensity decay is not a single exponential depending on material properties. In those cases an extended model of the intensity decay can be employed to assess all lifetimes present. The phosphorescence lifetime of the phosphor material decrease with increasing temperature, see Fig. 8a. The temperature can thus be calibrated to the lifetime or lifetimes and retrieved by calculating the phosphorescence lifetime(s) from the measured intensity decay. This is normally done by fitting the intensity decay to a theoretical model (e.g.

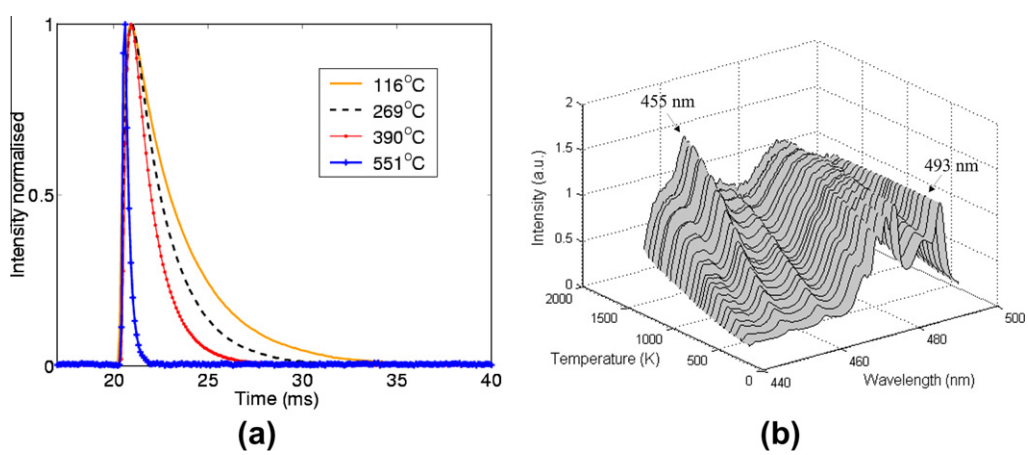

Fig. 8. TP-thermometry using (a) the temporal decay approach (YAG:Dy) and (b) the spectral method (Mg_3FGeO_4:Mn).

Eq. (2)), using a nonlinear fitting procedure. The error in temperature in such a measurement is ideally less than 1% [53].

When performing the calibration procedure, a single-exponential fit is often appropriate for phosphors that show multi-exponential features as well, i.e. the fit itself does not need to be perfect as long as it provides robust results.

In the spectral or intensity ratio measurement method illustrated in Fig. 8b, the fact that the various spectral emission regions for certain thermographic phosphors differ in how they are affected by temperature, is utilized. The intensity ratio of these two spectral positions indicates a temperature sensitivity that can be utilized for thermometry. The spectral method can be used for phosphors having two emission lines that are influenced differently by temperature variations, some phosphors have emission lines that are strongly broadened by temperature and others exhibit large line shifts. Some TPs and their temporal and spectral properties for thermometry are summarized in Fig. 9 [54]. As can be seen in most cases, the temporal approach normally gives a higher precision since the slopes of these curves may be steeper than the spectral ratio curves. Whereas the temporal approach has mostly been used for single point measurements, the spectral method can easily be extended to two-dimensional measurements [55–60]. For this purpose, two images corresponding to the emission originating from two different excited energy states need to be acquired simultaneously. One way of doing this is to use two ICCD detectors for registering emissions through use of a beam splitter and two band-pass filters. Here, a calibration measurement of the sensitivity of both ICCDs is needed to enable the pixel intensities in the different cameras to be balanced. Another method is to use an imaging stereoscope which uses mirrors to split the image into two identical images that can be recorded with a single camera. In recent years the development of temperature measurements using TP in connection to combustion/high temperature applications has increased dramatically. One area of great interest is to study flame spread scenarios. These phenomena have for long time been studied with the aim of improving theoretical models for flame growth and for the prediction of flame spread. Nevertheless, the complexity of the phenomena involved makes the development of a unified model or a set of such models a very slow process. One way of improving current models is to develop an experimental database, especially of surface temperature, which plays a major role in heat exchange between the flame and the solid material. For this purpose the fourth harmonic radiation of an Nd:YAG laser at 266 nm was used to excite the phosphor particles (Mg4FGeO6:Mn) deposited on/in the material of a low density fiber board (LDF) or Polymethylmethacrylate (PMMA). The temperature changes were measured when the material was exposed to flame spread by placing it in a pool fire scenario using alcohol or heptane as fuel. Both single point and two-dimensional experiments were successfully reported [61].

Related to fire studies is the pyrolysis of solid organic materials. Many parameters that play a key role in pyrolytic processes in the heat transfer models need to be validated and correctly assessed. Surface temperature is one of these parameters. In the work presented in Ref. [62], the surface temperature of decomposing construction materials, together with their mass-loss rates during pyrolysis were studied.

Thermographic phosphorescence has been used for measurements of surface temperatures in industrial apparatus, e.g. gas turbines and IC engines. In the work reported in [63], laser-induced phosphorescence thermometry has been applied to the inner wall of the afterburner on a full size fighter jet engine. The measurements were aiming at testing the technique under the extreme conditions prevailing in the test rig of an aviation gas turbine running at full afterburner load. The experimental set-up is shown in Fig. 10a.

The investigations were made on the outlet cone of the afterburner using the intensity ratio method, and the phosphor YAG:Dy, which is suitable for temperature measurement from room temperature to 1700 K, was used. A Nd:YAG laser running at its third harmonic, 355 nm, was used for excitation of the phosphor particles. A stereoscope containing interference filters for the wavelengths 455 and 493 nm was used for collection of signals emitted within the two spectral regions onto one half each of an image intensified CCD-chip. The intensity ratios of the resulting images were converted to temperatures, using predetermined calibration curves. Single-shot images of surface temperatures are shown for an afterburner of 100% load. The surface temperature at idling (no fuelling of the afterburner) was 517 K, whereas, as indicated in Fig. 10b, when the afterburner was activated the temperature increased and levelled off at temperatures around 940 K. After the engine was operated at full load with the afterburner for 50 s, the afterburner was deactivated and the temperature decreased. Statistics from the measurements show that at full afterburner load the standard deviation, or precision, was about 20–30 K at about 1000 K or 2–3%.

The use of thermographic phosphors in internal combustion engines allows both single-point and two-dimensional surface temperature measurements to be obtained. Thermocouples are a standard tool for temperature measurement. However, due to complex geometries and the harsh environment in engines thermocouples are time-consuming to install. In addition, once installed, thermocouples are difficult to shift to a different measurement area, which increases the complexity of an experiment and the time

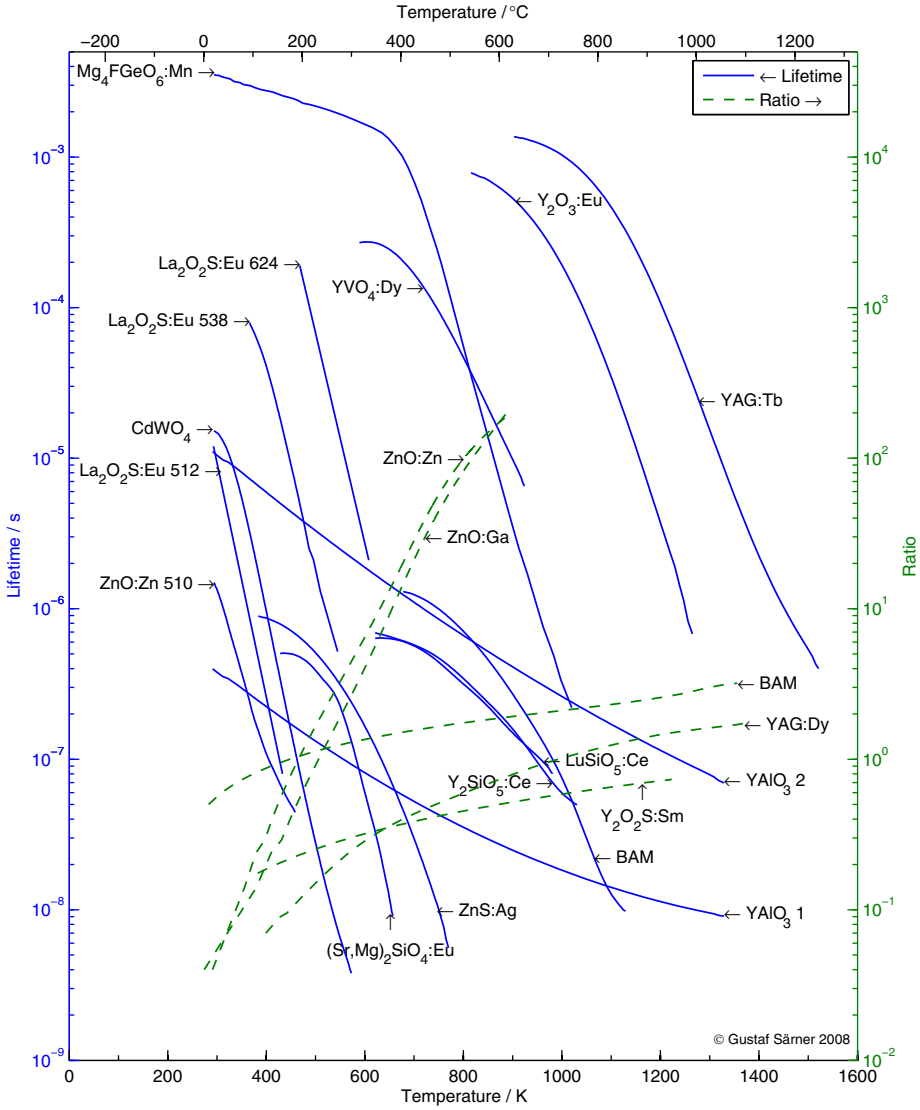

Fig. 9. Temperature sensitivity for some of the phosphors surveyed in the paper. Ratio calibrations are dashed and can be read on the *y*-axis at the right [54].

required for it. For moving targets, e.g. pistons and valves, the instrumentation of thermocouples is even more cumbersome. Thermographic phosphors also have the advantages of enabling two-dimensional measurements and of having a high degree of applicability to diverse geometries. The technique is also insensitive to engine vibrations or to typical pressures of around 30–100 bar inside the combustion chamber. Studies performed using either Cr^{3+} [64,65] or Eu [66,67] have not revealed any noticeable changes in lifetime or in spectral line positions at pressures under 1GPa. Armfield et al. [68] used TP for thermometry of the intake valve (on the stem side of the valve), and on the piston of an SI engine [69] both two-dimensional and one-point temperature measurements in a direct injection gasoline engine using TP are demonstrated and discussed. The measurements were performed on the intake and exhaust valves, and inside the combustion chamber on the pistons.

Recently the use of thermographic phosphors have been further developed not only for surface measurements but also for thermometry of droplets and in sprays [70,71] and even for measurements when seeding particles to a gas flow [72]. In the latter case TP was used for simultaneous PIV and phosphorescence, yielding both velocities

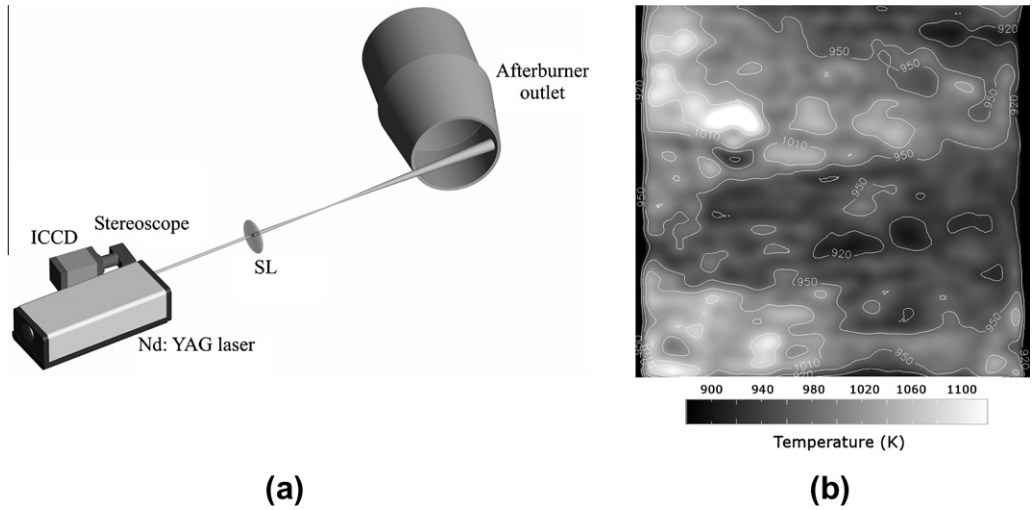

Fig. 10. (a) Experimental set-up for 2D surface thermometry in a afterburner to a full size jet engine. (b) 2D temperature map.

and temperatures. There has also been an effort to find phosphors with short phosphorescences lifetimes and with emission in the blue spectral region. This would facilitate measurement when there is a strong continuous emission background and also when fast moving objects are to be captured. A further desire has been to find phosphors with high spectral sensitivity for changes in temperatures, which would allow precise single shot 2D measurements. As was shown in Ref [73], ZnO:Zn and ZnO:Ga, have emission lifetimes on the nanosecond scale, emission in the blue spectral range and very strong spectral sensitivity to temperatures. This has been demonstrated by recording the ratio of two temperature-sensitive emission lines from burning droplets. A series of 2D temperature measurements have been performed on burning droplets of methanol [73]. A container with methanol and 1% ZnO:Ga was heated to the boiling point of methanol, inducing turbulence in the container and ensuring sufficient mixing of the insoluble inorganic phosphor and the methanol. The experimental set-up was similar to the one described in Fig. 10a. The droplets were ignited by falling through a Bunsen burner flame. The 5 ns long 355 nm excitation laser pulse was formed to a sheet which penetrated the center of the droplet. Figure 11 shows a temperature image of a burning droplet inside the Bunsen burner flame. The short exposure time and strong signal reduce the chemiluminescence interference to negligible intensities.

To date, basically all applications of TP thermometry have been based on commercially available phosphors with adequate spectral and/or temporal emission properties. An alternative and perhaps more appealing approach would be to use tailor made phosphors with characteristics optimized for each specific application. Such activities have started, e.g. at University of Duisburg where the host material, Al2O3, through CVD and the sol–gel method have extended its temperature range by multiple doping with Cr^{3+} and the rare-earth metals Dy and Tm [74].

4. Coherent techniques – Infrared Polarization Spectroscopy, IRPS

One of the first coherent nonlinear laser techniques to be used for combustion diagnostics was CARS, with applications in engines already in the late seventies [75]. Despite the attractive diagnostic feature of this technique, in that the signal is generated as a new laser beam, in contrast to the 4π sr. emission of LIF, this technique suffers from a relatively low sensitivity and experiments are mainly restricted to point measurements of major species. Thus, there has been a desire to combine the sensitivity and 2D capability of LIF with the coherent nature of CARS. Techniques which in principle provide these features are Polarization Spectroscopy, PS and Degenerate Four-Wave Mixing, DFWM. Both techniques are based on nonlinear optical interaction through the third order nonlinear susceptibility, $\chi^{(3)}$.

Most of the spatially resolved laser diagnostic applications for studies of combustion processes have been made in the UV/visible region of the spectrum by exciting electronic transitions, which is also true for PS. In this spectral region, PS has been widely used in combustion and plasma diagnostics, e.g. with single-photon excitation of OH

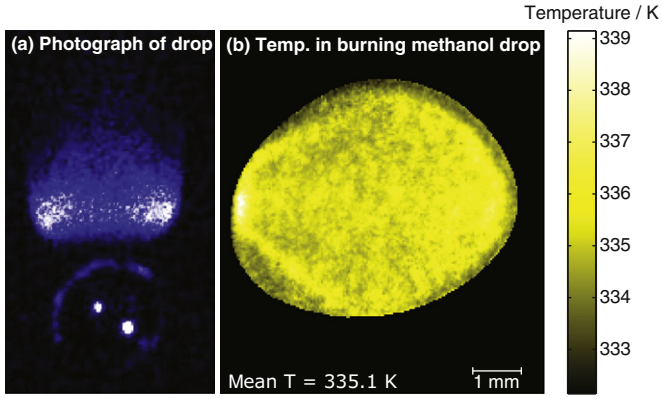

Fig. 11. (a) A burning methanol droplet as it was falling, photographed with an ICCD camera. (b) A temperature image of a burning methanol droplet that was falling, recorded in the middle of the Bunsen burner with a camera exposure time of 10 ns [73].

[5], C_2 [76], NH [77], NO [78] and with two-photon excitation of NH_3 [79], CO [79], N_2 [80,81]. Two-dimensional visualization using single-photon PS, on OH, e.g. [82], as well as two-photon PS on H [83] has also been demonstrated. Furthermore, comprehensive calculations based on direct numerical integration [84] have been performed aimed at quantitative detection with PS.

However, there are a number of species that for various reasons can not easily be measured in the UV/visible spectral region, e.g. CO_2, which however has been identified as yielding a potential background emission at high pressure [85], and CO, NH_3, H_2O, where, as described above, two-photon excitation schemes have to be used for flame detection using LIF, see e.g. [28,86,87]. In addition most hydrocarbon species of interest for combustion have very broad featureless UV spectroscopic signatures, which has created a strong need for spatial and temporally resolved detection of these species.

Probing molecular ro-vibrational transitions by IR excitation schemes has always been attractive to the combustion diagnostic community. Many important combustion species, which have no conveniently accessible electronic transitions, do have strong absorption in the mid-IR spectral range. As described, e.g. in the paper by Webber et al. [88], diode laser absorption measurements have been extensively and successfully used in the IR spectral regions for spatially averaged measurements in a variety of combustion applications. However, spatially resolved laser-based combustion diagnostic experiments in the mid-IR spectral region via ro-vibrational transitions have been limited by the relatively low IR fluorescence quantum yields. Although, there have been reports on using ro-vibrational transitions in the IR part of the spectrum for spatially resolved detection of some species, e.g. LIF detection of CO [89] and CO_2 [90,91].

Absorption-based coherent techniques with high spatial resolution, like PS and DFWM, can, however, be suitable alternatives to LIF in this spectral region. The strong coherent signal beam generated by these techniques also provides a sufficient discrimination against the background thermal IR radiation from hot reacting flows. IR-DFWM spectroscopy has been applied in the detection of C_2H_2 and CH_4 in low pressure chambers by probing the C–H asymmetric vibrations [92–94].

PS was firstly introduced by Wieman and Hänsch [95] as a Doppler-free spectroscopic method, related to saturation spectroscopy but offering a considerably higher signal-to-background ratio. A typical IRPS set-up is shown in Fig. 12. A strong pump beam and a weak probe beam, tuned to optical transitions of the target species with a common ground state or excited state, are crossed in the interrogation region. The optical pumping of the target species with the polarized pump beam produces birefringence and induces detectable polarization changes in the weak probe beam. In order to facilitate beam alignment, a He–Ne laser is spatially overlapped with the IR beam. The PS technique has the unique ability to distinguish spectral transitions belonging to P, R or Q branches by analyzing the J-dependence of the absorption cross-sections [96,97]. Being a coherent technique with a signal created as a laser-like beam, PS has obvious advantages for efficient collection of signal and discrimination against background interference from scattered light and flame emission.

Detection of CO_2 via IRPS has been investigated by probing rotational lines belonging to different overtone and combination bands [98,99]. In subsequent efforts, successful detection of CH_4 in an atmospheric pressure flow with a sensitivity of ~1000 ppm was demonstrated [100]. IRPS was also applied to other hydrocarbons. Shown in

Fig. 13 is an IRPS excitation scan of a gas mixture containing a small amount of CH_4 and C_2H_6 diluted with Ar [101]. These results have proven that spatially resolved rotational line resolved detection of a mixture of different hydrocarbon molecules is possible, although there is a manifold of overlapping lines.

The application of IRPS in flames obviously faces the challenges of a weaker signal due to the large partition function and a low number density at high temperature, as well as strong spectral interferences from hot water lines, etc. A thorough investigation of the application of IRPS to water and CO_2 molecules in flames was performed [102] at around 3700 cm^{-1}, where the strongest water band and the combination band of CO_2 exist. The strong interference of hot water lines, mostly from the Q branch lines in this spectral range, was found to be avoidable with a circularly polarized pump beam, which suppresses the Q branch lines. In a stoichiometric H_2/air flame seeded with CO, the CO_2 concentration in the burnt region can be well controlled by precisely adjusting the amount of seeded CO. A few thousand ppm of CO_2 in the hot burnt region was found to be detectable.

Detection of trace level HCl in combustion environments has been investigated using IRPS [103]. Shown in Fig. 14 are IRPS spectra obtained in an atmospheric pressure methane/air flame seeded with trace amount of chloroform, where all chloride atoms are converted to HCl. The black and blue-black curves are IRPS excitation scans recorded in flames with and without chloroform seeding. Due to the trace amount of chloroform seeded in the flame, the only difference between the two flames is the existence of a small amount of HCl in the seeded flame. The extra lines recorded in the chloroform seeded flame coincide well with a simulated HCl IRPS spectrum. The detection limit for flame conditions was estimated to be ~50 ppm. This result demonstrates the possibility for stand-off detection of toxic species with high spatial resolution.

The capacity of IRPS was further demonstrated by OH detection in a low pressure laminar H_2/O_2 flame [104]. These measurements are challenging mainly due to difficulties in spectral interpretation as the spectra are typically very congested and dominated by a large number of closely spaced hot water lines. There are usually quite a large number of hydrocarbon molecules and radicals in the combustion process as intermediate species while burning hydrocarbon fuels, especially in the reaction zone. In situ non-intrusive detection of these molecules and radicals will provide crucial information for the understanding of chemical details of the combustion process. IRPS, with its ability to provide spatially resolved probing of C–H stretching vibrations, could in principle take on this task. This possibility has been investigated in a low pressure methane/O_2 laminar flame [105]. Shown in Fig. 15 are spatially resolved IRPS excitation scans obtained at different heights above the burner surface. The upper trace (a) shows the C2H2 IRPS spectrum obtained in a gas sample of a cold flow. The lower trace (c) demonstrates the simulated IRPS spectrum of hot methane lines. The recorded IRPS spectrum shown in (b) was collected at different heights above the burner surface. It is clear that the spectral lines from methane are present. At 1 mm above the burner, the probed region is located in the preheat zone, where the methane lines are slightly weaker than the signal collected at 2 mm above the burner surface, where the temperature is higher due to the proximity to the reaction zone. Higher up in the flame, it can be seen that methane starts to be consumed. The production and subsequent consumption of acetylene can be clearly seen in the lines indicated by the red box of dashed lines. Also indicated in the figure are the methyl lines, highlighted by the brown box. Despite being a radical with rather weak absorption in this spectral range, it is clearly observable in the reaction zone.

The application of IR-DFWM in flames has been practically hindered by the challenges to align the IR beams due to the lack of proper beam locator in the mid-infrared spectral range. Recently, a stable beam alignment system has been successfully applied, which makes the application of IR-DFWM much more convenient [106] and preliminary results indicate that sensitivity of detection of HCl and C2H2 was similar to the IRPS results. Both IRPS and IR-DFWM hold much promise for quantitative non-intrusive measurement in flames especially for species like HCN and N_2O, which exist as stable species, providing easy on-line calibrations. Also the coexistence of multi-species in flames makes it possible for cross calibrations between different molecular or radical species.

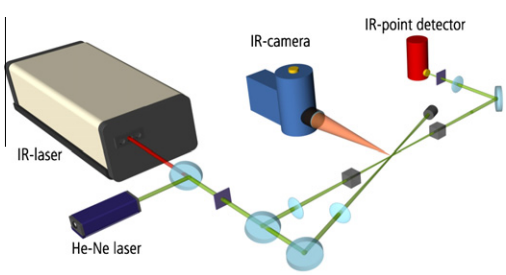

Fig. 12. Experimental set-up for IR Polarization Spectros copy.

5. Challenges and future applications

As indicated above the development of laser techniques for combustion diagnostics has been of utmost importance for a deepened understanding

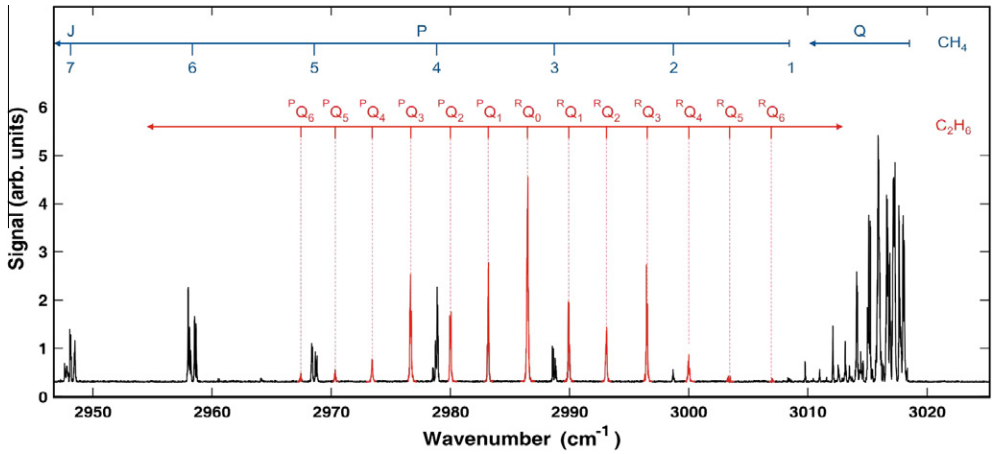

Fig. 13. IRPS excitation spectrum of a gas mixture containing CH_4 and C_2H_6 prepared in a 10 mm gas jet at ambient pressure and temperature. The sample gases were composed with 1.93% CH_4, 0.57 % C_2H_6 and 97.5% Ar.

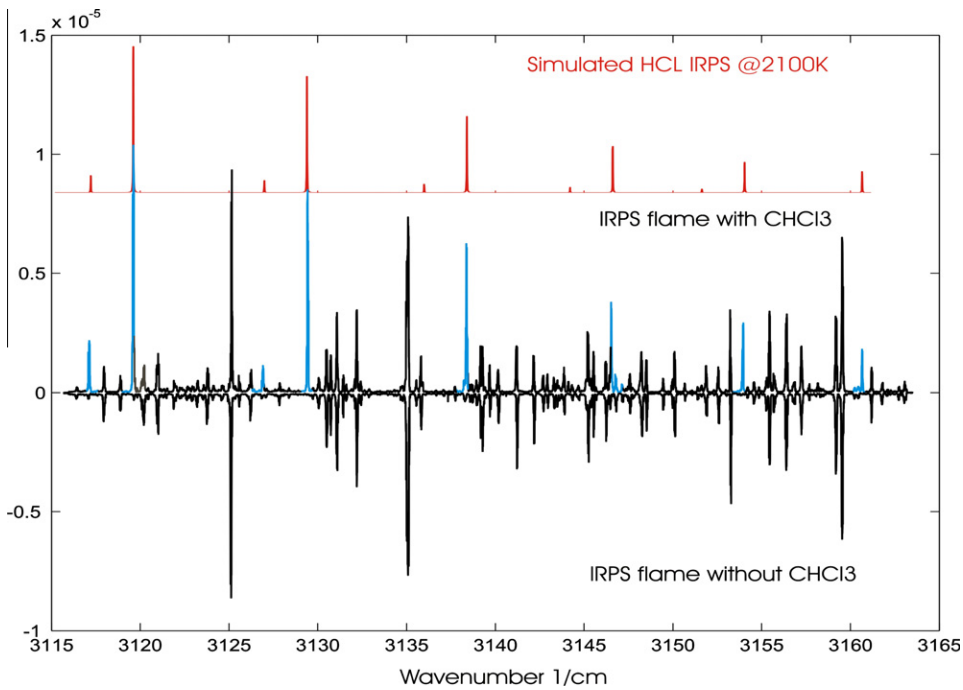

Fig. 14. IRPS excitation scans obtained in flames with (the middle curve) and without (the lower curve) chloroform seeding. A simulated IRPS spectrum of water lines at 2000 K in the same spectral range is also shown (upper curve).

of phenomena involved in combustion and thus for an optimum practical application. Still, there are several challenges where new ideas/approaches are needed. In this section some of these challenges and potential approaches to address them are outlined.

5.1. Limited optical access

In combustion research there is a strong need to measure inside devices with minimum optical access to avoid altering the combustion characteristics through the introduction of optical windows,

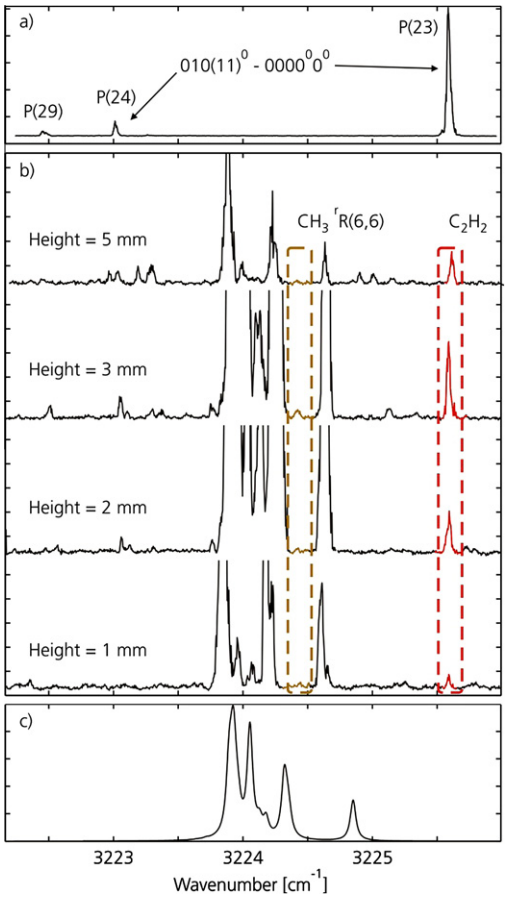

Fig. 15. IRPS acetylene spectra in cold flow and in a 50 mbar $\Phi = 1.5$ methane/oxygen flame. (a) Excitation scan over the P(24) and P(23) line of the $(010(11)^0)$–(0000^00^0) band of acetylene in an unburned gas flow containing acetylene (10 mbar, 0.3% C_2H_2 in N_2). (b) Excitation scan over different heights of the flame. The P(23) line from the $(010(11)^0)$–(0000^00^0) band of acetylene and the $^rR(6, 6)$ line of ν_3 band of methyl are indicated in the figure. The strong lines in the center part are from hot methane molecules. (c) Calculated IRPS spectrum of hot methane with molecular parameters taken from the HITRAN database.

e.g. the limited pressures reached in optical engines as compared to "steel engines". In terms of practical uses for thermographic phosphors, this technique has been used for piston thermometry in a production diesel engine using optical fibers for guiding laser light into the cylinder and optical fibers for collecting the phosphorescence to the detector [107]. With regard to LIF imaging in engines, measurements have been done using a conventional endoscope for fuel visualization [108]. Although applicable in several cases, this approach suffers from low detection efficiency and no UV sensitivity. However, as recently shown in Ref. [109] advanced LIF imaging with high optical throughput combined with good imaging quality has been developed and demonstrated in IC engines using specially designed optics causing minimal disturbances to the engine design. The endoscope developed for this application was found to provide comparable signal to a standard *f*/4.5 UV lens using the same image magnification. To further facilitate practical experiments in industrial environments, the authors also designed micro-optical endoscopes for laser beam shaping and delivery [109]. Figure 16a shows the experimental set-up in the engine while Fig. 16b shows the corresponding single shot images of fuel LIF using commercial fuel.

In addition to the need to do measurements in as close to "real" engines/gas turbines as possible, there is also a need for characterization of the burning conditions inside full-scale industrial combustors such as boilers in power plants or for studies of fires in order to maximize the efficiency and minimize emissions of pollutants and to improve extinguishing strategies in case of fire. Large boilers, however, represent really challenging measurement conditions, primarily because of their large size and limited optical access. For this purpose there have over the past few years been investigations of visualization techniques requiring optical access only in one direction, i.e. the so-called single-ended visualization techniques. These techniques are based on structured illumination and picosecond LIDAR, respectively.

5.1.1. Structured illumination

The main difficulty of achieving a 2D image with only one optical access, i.e. when only back-scattered light is studied, is due to the fact that light originating from the parts of the sample which are not within the depth of field of the imaging system, the so-called out-of-focus light, will have a blurring effect on the resulting image. One approach to reduce the intensity of this light is to use a so-called obscuration disk [110]. Here, the center of the imaging lens is covered in order to discriminate against photons not having a certain incoming angle. Now, only photons originating from within the depth of field of the imaging system will be imaged properly. Unfortunately, this technique is only applicable for point measurements. The reason for this is that in two dimensions, out-of-focus photons which originate from parts of the sample lying off the optical axis may now have the same incoming angle as those originating from the in-focus part of the sample. In microscopy, where single optical access is a normal approach, another technique, known as structured illumination, has been developed for discrimination against the out-of-focus light [111]. With this technique it is possible to separate the light coming from the in-focus plane of the sample from all other detected light, through the so-called optical sectioning. This is made possible by using spatially modulated illumination and is based

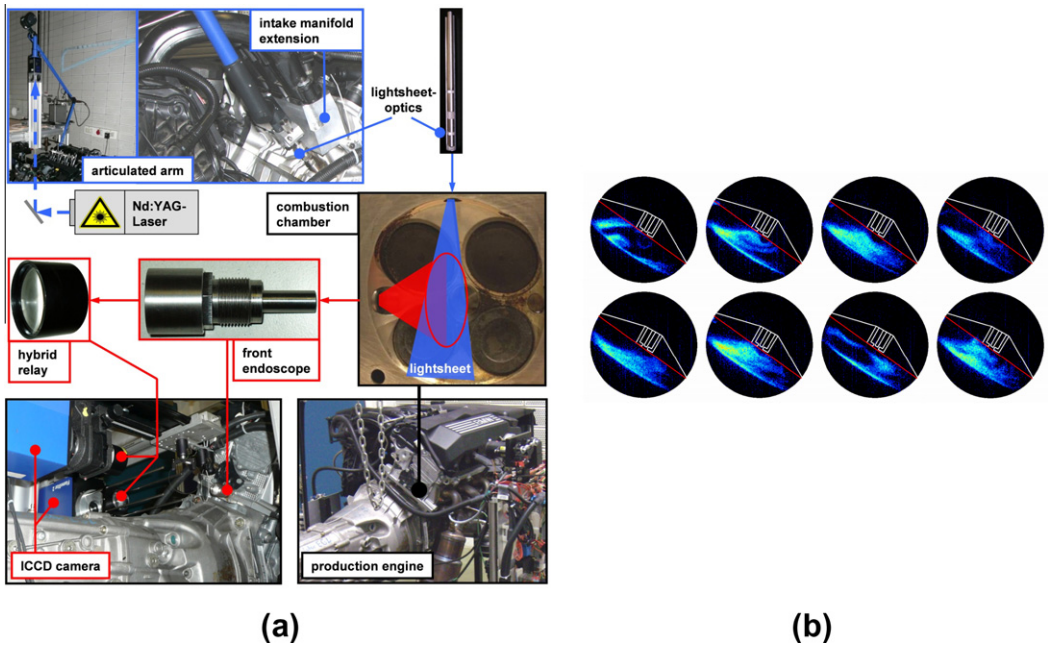

Fig. 16. Endoscopic LIF measurements in an engine: (a) the experimental set-up and (b) single shot LIF images at 5 CA ATDC. (Courtesy of Gessenhardt, Zimmermann, Schultz, Reichle, Pruss and Osten [109]).

on the fact that only the zero spatial frequency is not attenuated by defocusing. Any spatial modulation of the excitation light will therefore only be visible in the in-focus plane of the sample.

Hence, the main idea is to record three images, I_1, I_2, and I_3 where the intensity modulation is shifted a third of a period ($2\pi/3$) between the images. It can be shown that I_S represents an image where the out-of-focus light is removed [111];

$$I_s = \frac{\sqrt{2}}{3}\left[(I_1 - I_2)^2 + (I_1 - I_3)^2 + (I_2 - I_3)^2\right]^{1/2} \tag{3}$$

whereas I_C constitutes the conventional image;

$$I_C = \frac{I_1 + I_2 I_3}{3} \tag{4}$$

The sectioning strength (depth-resolution) depends mainly on two factors: the frequency of the modulation and the depth of field of the imaging system, and can be determined by measuring the modulation depth while moving a reflecting test object through the focus [112].

In order to validate whether optical sectioning is possible for single ended visualization of non-solid objects, a flow of water droplets generated by a nebulizer was studied [113]. An internal structure, which differs from the water droplets in terms of reflectivity, was introduced into the sample. The aim of this was to optically select the layer where the internal structure, which in this case was a flow of nitrogen, was visible. The two flows were assumed to be sufficiently stable and separated from each other close to the outlet of the nebulizer, while, due to fluid dynamics, mixing between the flow of water and nitrogen will occur higher up.

In Fig. 17a and d 1 of the 3 spatially modulated images are shown for 1 and 20 accumulations, respectively. Note that one accumulation simply means that only one set of three spatially modulated images (I_1, I_2 and I_3) was used, while for 20 accumulations, 20 sets were averaged. As described above, both the conventional image and the sectioned image are recorded simultaneously, by either summation (Eq. (4)), which gives the conventional image, or by studying the differences of the images I_1, I_2 and I_3 (Eq. (3)), resulting in the sectioned image. The resulting conventional images are shown in Fig. 17b and e. Here, due to out-of-focus light reaching the detector, the internal structure is smeared out. The out-of-focus light is efficiently removed by the use of structured illumination, shown in Fig. 17c and f, with the same number of accumulations as indicated for Fig. 17a and d. Here, the internal structure created by the flow of nitrogen is clearly visible in both sectioned images.

5.1.2. Picosecond LIDAR

In combustion research, most laser diagnostic measurements have been limited to small and moderately sized combustion devices, such as

laboratory flames, internal combustion engines, and gas turbines, see, e.g. Ref. [16] and references therein. Nevertheless, laser diagnostics using sensors based on tunable diode laser absorption spectroscopy (TDLAS) have been demonstrated in full-scale power plants for temperature and species detection, see Ref. [114] and references therein. However, since TDLAS sensors are line-of-sight integration devices, they normally do not provide spatially resolved information.

Light detection and ranging (LIDAR) is a well established method for remote detection and ranging of objects and gaseous plumes. The fundamental principles of LIDAR are described in a number of textbooks, see, e.g. Measures [115]. Using laser pulses of 30 ps duration, instead of pulses of 5–10 ns duration which are typical for standard LIDAR experiments, the range resolution can be improved from ~m ranges to ~mm ranges. Thus, the ps-LIDAR technique has a

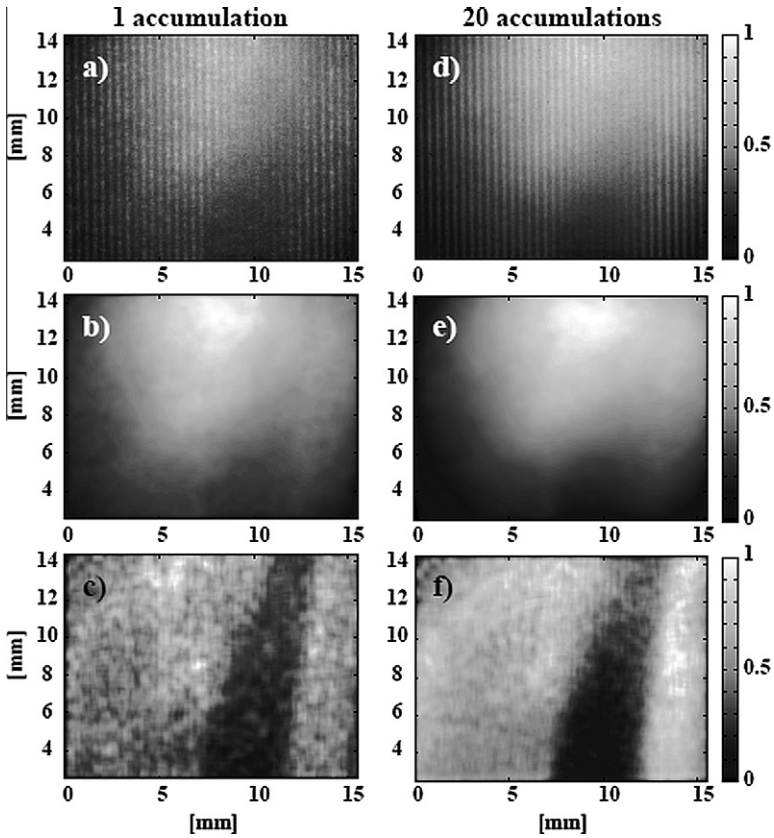

Fig. 17. In (a) and (d) I_1 is shown for 1 and 20 accumulations, respectively, where the superimposed grating pattern can be seen. (b) and (e) shows the conventional back-scattered images for the same number of accumulations. The internal structure created by the flow of nitrogen is extremely smeared out in both of these cases. When removing the out-of-focus light, by the use of structured illumination, the internal structure is clearly visible, shown in (c) and (f).

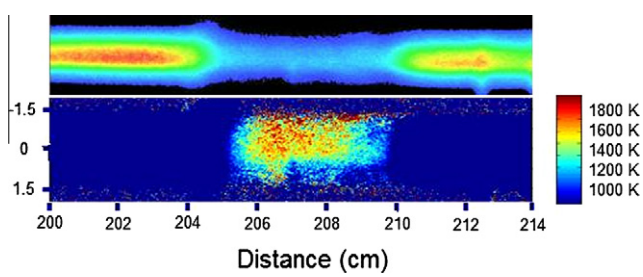

Fig. 18. Rayleigh thermometry with ps-LIDAR in an ethylene/air flame on a McKenna burner. Recorded Rayleigh intensity (upper part) and evaluated temperature image (lower part).

unique capacity to measure over a long distance (meters) yet providing good spatial resolution (mm). In addition, LIDAR requires only one optical access, i.e. only one window is needed on the combustion device to be studied. These features make the technique attractive for a number of applications in full-scale combustion devices; for example studies of particle formation and distribution using elastic scattering, thermometry using Rayleigh scattering (in non-sooty environments), and species concentration measurements using differential absorption LIDAR (DIAL).

Using a set-up based on a mode-locked Nd:YAG laser (Ekspla, PL 2143C) and a streak camera (Optronis, Optoscope) the potential for ps-LIDAR as a large-scale combustion diagnostic has been evaluated. Experimental details are described in [116]. Results from measurements in a non-sooty Bunsen flame indicate that ps-LIDAR might be applicable for single-shot Rayleigh thermometry, possibly yielding two-dimensional information, in combustion devices with insignificant particle concentrations. The corresponding results from a sooty flame show that the technique also could be used for qualitative soot distribution measurements in particle environments. Figure 18 shows an example of a Rayleigh-temperature image recorded with ps-LIDAR in a stoichiometric ethylene/air flame on a McKenna burner at a distance of $\sim$2 m. In this type of flame ps-LIDAR Rayleigh thermometry shows good agreement with CARS measurements up to $\varphi = 1.3$ (fuel equivalence ratio). The uncertainty for two-dimensional thermometry has been found to be less than ± 60 K. Furthermore, results from DIAL measurements in acetone jets indicate a good potential for using DIAL for fuel visualization. Overall, the results are promising with respect to future applications in large-scale combustion devices.

5.2. Measurements in optically dense media

As described above planar imaging techniques have been decisive for a deepened understanding of combustion/flow phenomena. For spray research several techniques have been applied, e.g. Particle Image Velocimetry [117], Planar Laser-Induced Fluorescence [118], and Laser Sheet Dropsizing [119]. Although these techniques use different properties of light scattering and instrumental approaches, they are all based on the single scattering approximation; assuming that the detected photons have experienced only one scattering event prior to arrival at the detector. This assumption remains valid when the number density of particles is low and when the total photon path length within the probed medium is short. However, within optically dense media (optical depth $\geqslant 1$), such as the region near the nozzle exit of a spray injector, a large amount of photons are scattered more than once and the single scattering assumption is no longer valid. In this case, the multiple light scattering (also known in the literature as secondary emission) blurs and attenuates the recorded images, introducing significant uncertainties in the detected optical signal. This contribution from multiply-scattered photons is particularly important in planar imaging, due to the use of both a relatively wide source of light and a large detection acceptance angle of the collection optics. Due to the complexity in predicting and understanding effects introduced by multiple scattering, no experimental technique has been proposed, so far, for the correction of the secondary emission in laser sheet imaging. Only two other line-of-sight imaging techniques, in which multiple scattering is mitigated, currently exist; Ballistic Imaging (BI), e.g. see [120] for an updated review in this field, and X-ray Imaging, e.g. see [121] for application in a gasoline injection (GDI) nozzle. Both techniques have distinct advantages but are experimentally rather complex, especially X-ray imaging, which requires a synchrotron light source.

Very recently it has been shown that a similar technique, based on structured illumination, as described above for single ended experiments, has a great potential to suppress the multiple scattering effects in laser sheet imaging of sprays, and can be seen as a complement to BI in moderately turbid media (optical depth < 8). The technique, Structured Laser Illumination Planar Imaging, SLIPI, is similar to the work first published by Breuninger et al. known as Single Plane Illumination Microscopy – Structured Illumination (SPIM-SI) [122] which concerned the study of static solid objects at microscopic scale.

The technique is based on planar laser imaging but where the spatial sinusoidal intensity modulated laser sheet is created as above using a transmission grating. In the same way as out-of-focus scattering was reduced, directly scattered light can be distinguished from the multiple scattered light, since only single scattered light keeps the sinusoidal information contained within the laser sheet. Light that has experienced several scattering events prior to detection will instead appear as a non-modulated dc-offset in the recorded image. By recording the three images, where the modulation is shifted one-third of a period, this unwanted contribution can be mitigated in the image post-processing. Multiple scattered light is unaffected by this shift and therefore appears nearly identical in these three images. In contrast to multiply-scattered light, singly scattered photons will possess unique image features, due to the induced phase shift. By studying the absolute differences between the images, the multiple scattered light will be highly suppressed and the resulting image will mainly be built up by directly scattered light, resulting in more accurate imaging and thus improving the spray analysis.

In the work reported in Ref. [123], the probed medium was a hollow-cone water spray operating in the atomization regime. Particular attention was given to the dense spray region where conventional methods are hampered by errors due to multiple scattering. A quantitative estimation of the image improvement was performed based upon direct comparison between the conventional and the SLIPI image in studies of a hollow cone spray based on recording an average of 50 single SLIPI images as can be seen in Fig. 19.

More recent work has included the extension of SLIPI also to single shot experiments in fast and ultra fast applications (i.e. 100 μs and 100 ns, respectively) [124,125].

5.3. New species detection

Although many combustion species are detectable using various laser techniques, there are still several important molecules which have been difficult to measure using these techniques. The detection of two of these species is demonstrated using photofragmentation laser-induced fluorescence.

5.3.1. Photofragmentaion laser induced fluorescence (PF-LIF)

Many molecules have only weakly or non-fluorescing excited electronic states because of a large degree of internal energy transfer or because the excited state is unbound or has a very long lifetime. Two examples of species that do not fluoresce upon electronic excitation are the hydroperoxy radical (HO_2) and hydrogen peroxide (H_2O_2). These species are crucial intermediates in various oxidation processes, e.g. in atmospheric chemistry, combustion, and plasma chemistry. The branching chain oxidation of H_2 with O_2, forming HO_2 and H, is one of the most studied reactions in combustion. Michael et al. [126] have found that this reaction is the most probable initiation process in high-temperature oxidation of H_2. Reactions involving HO_2 and H_2O_2 are pivotal in high-pressure autoignition initiating engine knock [127]. In homogeneous charge compression ignition (HCCI) engines, which rely on autoignition, the decomposition of H_2O_2 into two OH radicals has been identified as the triggering ignition mechanism [128]. Hence there is a great demand for techniques that can detect hydrogen peroxides in situ without perturbing flow fields and temperature distributions.

Although HO_2 and H_2O_2 cannot be detected directly with laser-induced fluorescence (LIF) they can be detected indirectly by probing the OH fragments generated upon photodissociation. Probing the OH fragments with LIF provides high sensitivity and facilitates two-dimensional imaging. This pump–probe technique, called photofragmentation laser-induced fluorescence (PF-LIF), requires two laser pulses; a pulse for photodissociation (pump), typically in the UV regime, and a laser pulse whose wavelength is tuned to an absorption line of OH (probe).

Ever since Raiche et al. [129] suggested that photofragmentation fluorescence could be a useful diagnostic for acetylene detection in flames, various laser-fragmentation techniques have been used for combustion studies. For example, excimer laser fragmentation-fluorescence spectroscopy (ELFFS) has been used to detect alkali species in pressurized boilers [130,131] and for detection of particulate matter in Diesel generator engine [132]. Furthermore, laser-induced fragmentation fluorescence (LIFF) from C_2-radicals has been used for detection of vinyl radicals and acetylene [133]. Concerning hydrogen peroxides, PF-LIF has been demonstrated for HO_2 detection in a nitrogen bath gas (300 Torr) at room temperature [134]. Johansson et al. [135] have demonstrated two-dimensional PF-LIF measurements of gas-phase H_2O_2 concentrations in room-temperature mixtures of H_2O_2, water, and air at atmospheric pressure.

Recently, Li et al. [136] have demonstrated experimental 2D-visualizations of hydrogen peroxides in a flame. The species were detected in a Bunsen-type premixed methane/air flame. The fourth harmonic radiation (266 nm) of a pulsed Nd:YAG laser was used for photolysis, while the generated OH fragments were probed through their fluorescence induced by a probe laser tuned to the $Q_1(5.5)$ transition at ~283 nm. The probe laser beam was generated by a frequency-doubled dye laser (Rh590 dye) pumped by the second-harmonic radiation (532 nm) of an Nd:YAG laser. The PF-LIF signal was detected with an ICCD camera. More experimental details are found in [136]. Typical results are shown in Fig. 20. The signal due to OH radicals naturally occurring in the flame, recorded by having the photolysis laser blocked, was subtracted, providing an image that exclusively reflects the concentration of OH fragments generated by photolysis, and hence the presence of H_2O_2 and HO_2, as shown in Fig. 20b. A horizontal cross-section of the image shows two distinct peaks, symmetrically located on either side of the flame center (see Fig. 20c), stemming from the hydrogen peroxides.

5.3.2. Visualization of EGR (exhaust gas recirculation) gases

For conventional IC engine combustion, the fact that some burned gases are left in the cylinder to participate in (and thereby influence) the combustion of the following cycle have called for a diagnostic tool to visualize the distributions of these species. Furthermore, recent Diesel engine running strategies rely on the use of excess amounts of EGR to increase the ignition delay in order to decrease the temperature and thereby reduce both soot and

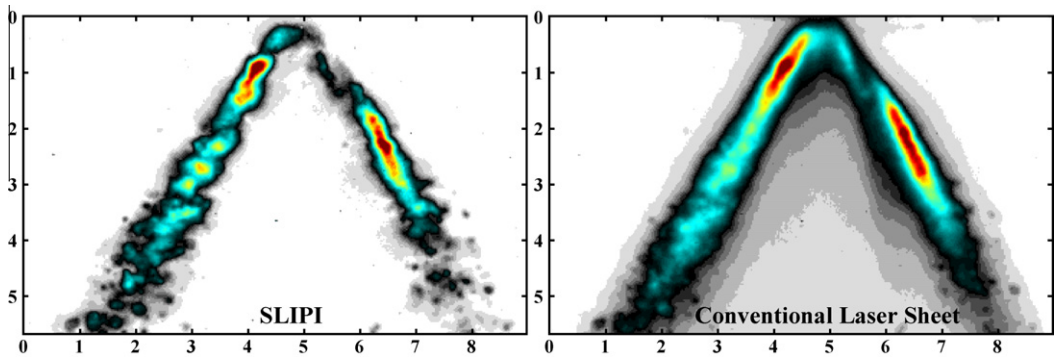

Fig. 19. Comparison between SLIPI and conventional laser sheet imaging of a hollow-cone water spray.

NO_x formation. Other novel concepts like CAI (an HCCI derivative) uses negative valve overlap to deliberately capture internal EGR in order to control the ignition timing by governing the in-cylinder temperature and dilution. In a multi-cylinder engine it is not only the total EGR ratio that is important. It is also crucial to achieve a flow that is balanced between the cylinders over the entire load/speed range. It is thus of crucial importance to visualize/measure gases representing the exhaust gases in IC engines, as well as in other combustion apparatus.

By investigating some of the main components in the exhaust gas it can be seen that there are several candidate species which could be considered to visualize EGR.

Nitric oxide detection by excitation in the UV spectral region is possible. However, this compound is not an attractive residual gas indicator, since the production of NO is a strong function of temperature. Cycle-to-cycle variations in combustion will therefore lead to large variations in NO production. Thus, the amount of NO is not well-correlated to the total amount of exhaust gas. In other words, increased production of NO in one cycle will lead to an overestimation of the residual gas concentration in the following cycle. Still, this technique has been used for visualization of residual gases at the time of ignition [137]

Nitrogen dioxide has an even more doubtful potential than nitric oxide. First, it is produced in small quantities. Second, the detection limit at high pressures is uncertain due to the long lifetimes of the excited states. Finally, the NO_2 concentration exhibits large cycle-to-cycle variations, like those of NO mentioned above.

LIF visualization of CO, H_2O and H_2 all rely on two photon excitation processes, and although both CO, H_2O have been detected in engines, the latter even for EGR studies, the high pressure coupled with the low concentration at lean conditions (CO/H_2) and the mixing of fresh air and residuals (H_2O) currently makes these species challenging to use.

Carbon dioxide seems to be an ideal species from a combustion point of view but due to the lack of proper absorption transitions in the UV and a low fluorescence yield in combination with problems of background suppression in the IR spectral region, no EGR measurements in engines using this species have been reported. In high pressure flames, CO_2 has been observed to absorb UV emission, which may disturb NO PLIF measurements [138,139] in engine applications. Although at room temperature, CO_2 is transparent to UV radiation at wavelengths greater than 205 nm, at flame temperatures it absorbs UV radiation between 200 and 320 nm. As a major product in hydrocarbon fuel combustion, the dependence of CO_2 absorption in the UV on temperature has been utilized to infer temperature in high pressure flames [140]. UV laser-induced fluorescence has also been observed in high pressure flames [141,142], however, no report of CO_2 UV LIF for EGR visualization has been reported so far for due to the small fluorescence quantum yield and absorption cross-sections [143].

Thus, at present, there are no real good candidates for direct visualization of EGR. Raman scattering is an excellent candidate for point measurements, but for single shot 2D measurements even the elevated pressure in an engine this is a challenge. However, recent work shows indirect (negative-PLIF) 2D imaging of EGR in an IC engine utilizing 3-pentanone as a tracer [144].

5.4. Quantitative measurements

As shown in this and many other diagnostic papers, the visualization of combustion parameters, mainly species concentrations and temperatures, has been a key element for a deepened understanding of combustion phenomena. Still it is a fact that almost all of the species visualization images published in the literature are of a qualitative nature. In the case of LIF the main difficulties for quantitative measurements are, as described in the introduction of Section 2, the problems associ-

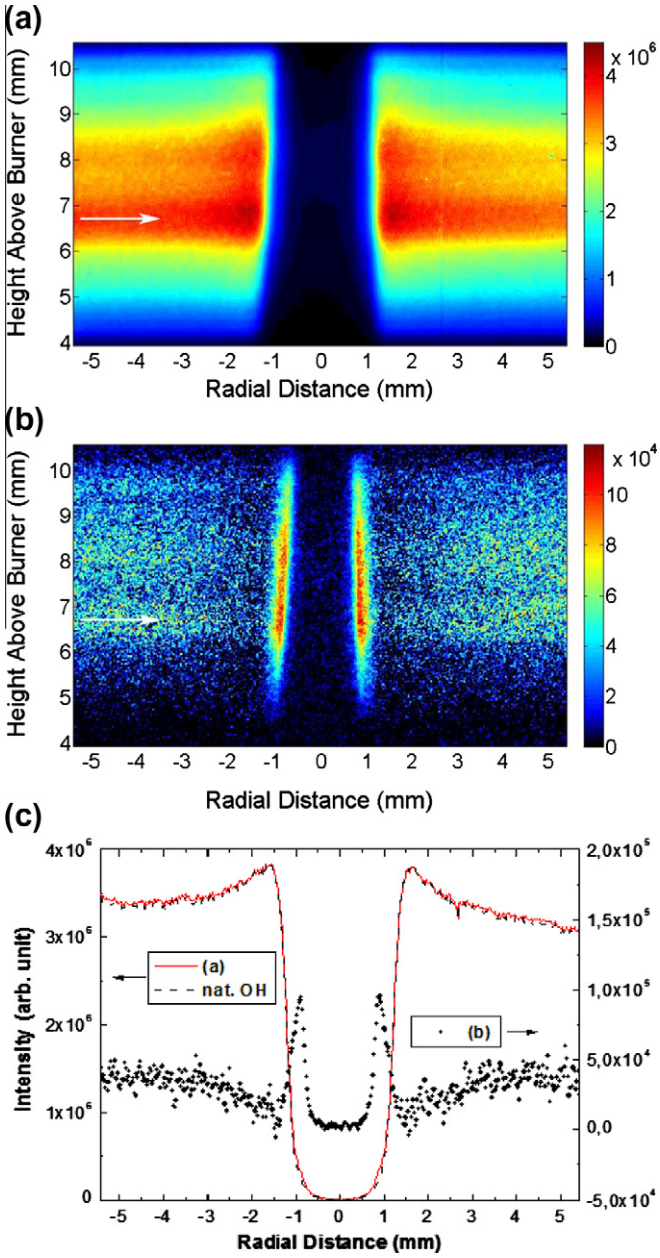

Fig. 20. (a) Typical PF-LIF image (1500 accumulated recordings) achieved with both photolysis and probe lasers fired. (b) Image after subtraction of signal from natural OH. Cross-section plots of the images shown in (a) and (b), taken at the vertical position indicated by the white arrows, along with the corresponding cross-section of the image reflecting natural OH.

ated with quenching corrections. Clearly, this is a formidable task, and approaches to handle this, e.g. saturation, suffer from spatial, temporal and frequency wing effects [15]. Quantitative 2D LIF measurements using in situ absorption calibration has been presented for OH [145] but has a limited dynamic range and has not been demonstrated for other species. Actual measurements of the quenching corrections require picosecond excitation and detection and have been demonstrated for point measurements, e.g. [146] whereas 2D measurements have been sparse [147,148]. In the recently developed method by Ehn et al. [148] fluorescence lifetime images are created based on picosecond laser excitation and detection with two ICCD cameras having a short and a long

gate, respectively. The camera having the short gate detects the early part of the fluorescence whereas the camera with the long gate records the entire fluorescence. The ratio between the fluorescence intensity registered with the short-gate camera and the intensity registered with the long-gate camera is a function of the fluorescence lifetime. Hence, from each pair of corresponding pixels an intensity ratio can be determined. Based on calculated fluorescence decay curves with lifetimes ranging from hundreds of picoseconds to tens of nanoseconds and carefully measured camera gate functions, a function describing lifetime vs. intensity ratio is extracted. Using this function, the measured intensity ratios are converted into lifetimes, resulting in a 2D lifetime image. This image can be converted into a fluorescence quantum yield image, which then can be used to compensate the PLIF image recorded with the long gate for fluorescence quantum yield variations.

For single shot temperature visualization, there have been several demonstrations, see, e.g. [16], however, the precision reached is not close to what for example a point wise CARS measurements can show. In order to meet the needs of modelers, it is thus important that the accuracy and precision in single shot 2D thermometry be increased.

Clearly there are further diagnostic challenges for laser diagnostics in combustion. One area where developments are needed is high precision velocity measurements without the need to seed the flow with particles, i.e. using molecules which can be easily seeded and do follow the highest fluid motions. At high flow speeds, the use of the velocity dependent Doppler shift of the absorption line of a species can be used, which was demonstrated already in the eighties, e.g. [149,150]. However, at low flow velocities, ~tens of meters per second, the precision of this technique is too low. Flow tagging is another technique which has been used for velocity measurements, see, e.g. Ref. [151] for a recent work in this field. 2D measurements with high spatial resolution remain a challenge.

Another field which needs further attention is quantitative fuel visualization when practical fuels are used, e.g. gasoline, diesel or aero fuels. A conventional way to visualize a fuel is to seed a fluorescing species to a non-fluorescing one component fuel, e.g. 3-pentanone and iso-octane. However, if and when a practical fuel, consisting of a "soup" of several individual hydrocarbon species has to be used, quantification is a much more challenging task.

5.5. Measurements in environments possessing strong background radiation

A challenge in many combustion diagnostics situations is to extract information when there is a strong laser-induced background emission, e.g. Mie scattering from particles and/or laser-induced fluorescence interferences from large molecules, which may be many orders of magnitude stronger than the signal we want to extract. Below are two approaches which may be useful in this respect.

5.5.1. Filtered Rayleigh scattering

Laser-induced Rayleigh scattering (LRS) is an outstanding diagnostic tool due to its simplicity and ease of implementation for the study of gases in combustion environments [152,153]. Due to the relatively strong scattering cross-section of Rayleigh scattering (which is generally 1000 times stronger than Raman scattering), LRS has shown its utility in quantitative 2D gas density or temperature measurements. However, for the elastic nature of Rayleigh scattering, it is often difficult to decouple LRS from any spurious elastic background such as Mie scattering from particles and scattering from optical windows. Atomic or molecular filters, made of an optical cell filled with atomic or molecular gas provides ultra narrow and strong absorption lines, which can be used to spectrally separate the un-broadened spurious light from part of the thermally broadened molecular Rayleigh scattering. This technique, named filtered Rayleigh scattering (FRS), was applied by Miles et al. in 1992 [154] in their study of a supersonic flow. For the broadly available injection seeded Nd:YAG lasers and the easy handling of I_2 molecules, two-dimensional FRS temperature measurements have been performed by a few groups combining the 532 nm laser and molecular I_2 filter. With this technique, Hoffman et al. [155] performed two-dimensional temperature images in a slightly sooty CH_4/air flame. Simultaneous 2D FRS temperature imaging and particle image velocimetry (PIV) measurements were performed by Elliott et al. [156] in a methane/air stagnation-flow flame and by Most et al. [157] in a turbulent flame. Gas temperature measurements using ultraviolet FRS, based on an atomic mercury filter (absorption line at 254 nm) together with a narrow linewidth Ti:Sapphire laser, has been performed by Yalin and Miles [158]. Yalin et al. [159] have also demonstrated this approach for thermometry in plasmas. Recently, 2D thermometry measurements in non-premixed flames were performed by Kearney et al. [160]. As an atomic filter, the mercury line is superior to I_2 lines, exhibiting a much sharper absorption notch, which enables a stronger tolerance of the scattering light intensities, which has been further proven by Zetterberg et al. [161] combining an isotope-enhanced mercury filter with a single-mode alexandrite laser [19]. However, for a general purpose tool of temperature

imaging with FRS, knowledge of molecular Rayleigh scattering line shapes at different pressures are required.

5.5.2. Temporal filtering of LIF/Raman measurements

A temporal filtering technique, complementary to spectral filtering, has been developed for laser-diagnostic measurements. The filter is applicable in cases where the interfering signal and the signal of interest have different temporal characteristics, for example LIF disturbed by elastic scattering and Raman scattering, i.e. inelastic scattering, perturbed by fluorescence. For the interfering-signal discrimination a picosecond laser system along with a fast time-gated intensified CCD camera are used. The concept has been demonstrated for suppression of interferences in both PLIF [162] and Raman measurements [163,164]. In the PLIF measurements the temporal filter was used to suppress laser-induced luminescence from a metal surface and to reject Mie scattering from water droplets. The results indicate that temporal filtering is a tool that can be very useful for PLIF diagnostics near surfaces and in environments where strong elastic scattering is present.

Interfering fluorescence is also often a main obstacle in Raman spectroscopy. Fluorescence can be suppressed with temporal filtering, taking advantage of the fact that Raman scattering is essentially instantaneous whereas fluorescence has a finite lifetime, typically on the nanosecond scale. In the set-up developed by Ehn et al. [164] a mode-locked Nd:YAG laser, providing laser pulses of 30 ps duration, is used together with either a streak camera or an ICCD camera with a fast gate for detection. Efficient suppression of interfering fluorescence by temporal filtering has been successfully demonstrated. It has been shown that this approach has the capacity to uncover several Raman peaks that, using conventional time-integrated detection, would have been completely drowned in fluorescence. For a model case of nitromethane contaminated by Rhodamine 590 dye it was found that the temporal filter improves the signal-to-interference ratio by a factor $\sim$20 in comparison with standard detection [164]. The concept has also been demonstrated for planar Raman imaging, using 266 nm excitation, of a levitated water droplet surrounded by nitrogen/air seeded with toluene [165]. The results clearly indicate that temporal filtering can be an attractive complement to spectral filtering for imaging as well, which could be particularly useful in diagnostics of two-phase systems.

An area which may also need further attention includes diagnostics near and on surfaces where strong scattering from the surface may cause spectral interferences. Another field where the importance of laser diagnostic techniques has a high potential for growth is in the diagnostics of plasma combustion, where the abundance of ions, excited species and non-thermal equilibrium conditions make these techniques invaluable tools.

6. Concluding remarks

In terms of understanding combustion phenomena, it is clear that besides the introduction and use of fast computers for simulations, the application of various laser diagnostic techniques have been the most important development during the last few decades. The techniques and their application, divided here into three categories, are all very important. For example in order to use the techniques for a deepened understanding of IC engine combustion, it is of crucial importance that proper laboratory investigations have been made in order to address the challenges encountered at high pressure and temperature, e.g. linewidth and collisional effects, spectral interferences as well as potential effects induced by the (high) laser power, e.g. breakdown, photolysis, Stark effects, incandescence, and scattering.

In terms of an optimum use of laser diagnostic techniques, it is also of crucial importance that a close connection is created with modeling activities, e.g. kinetics and/or fluid mechanics. Clearly, a prerequisite for such collaborations is a mutual appreciation and understanding of the respective fields involving diagnosticians, modelers and engineers.

As described above, there are still outstanding challenges in terms of applying laser diagnostic tools for quantitative visualization measurements in combustion environments. It is obvious that the fast development of the laser and detector technology will facilitate the solution of these challenges, while new ideas and smart approaches will still be leading this development.

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